

Application of Eq. (9.124) specifies the magnitude of the incremental length l :

$$l = \frac{n_0 u_p}{f_0} = \frac{9 \times 0.8 \times 3 \times 10^8}{10^{10}} = 21.6 \text{ cm}.$$

In summary, with $N = 6$ and $kd = \pi$, Eq. (9.121) becomes

$$F_{\text{an}}(\gamma') = \frac{\sin^2(3\gamma')}{36 \sin^2(\gamma'/2)}$$

with

$$\gamma' = kd(\cos \theta - \cos \theta_0) = \pi(\cos \theta - \cos \theta_0)$$

and

$$\cos \theta_0 = \frac{2n_0\pi}{kd} \left(\frac{\Delta f}{f_0} \right) = 18 \left(\frac{f - 10 \text{ GHz}}{10 \text{ GHz}} \right). \quad (9.130)$$

The shape of the array pattern is similar to that shown in **Fig. 9-30**, and its main-beam direction is along $\theta = \theta_0$. For $f = f_0 = 10 \text{ GHz}$, $\theta_0 = 90^\circ$ (broadside direction); for $f = 10.48 \text{ GHz}$, $\theta_0 = 30^\circ$; and for $f = 9.52 \text{ GHz}$, $\theta_0 = 150^\circ$.

For any other value of θ_0 between 30° and 150° , Eq. (9.130) provides the means for calculating the required value of the oscillator frequency f .

Concept Question 9-11: Why are antenna arrays useful? Give examples of typical applications.

Concept Question 9-12: Explain how the pattern multiplication principle is used to compute the radiation pattern of an antenna array.

Concept Question 9-13: For a linear array, what roles do the array amplitudes and phases play?

Concept Question 9-14: Explain how electronic beam steering is accomplished.

Concept Question 9-15: Why is frequency scanning an attractive technique for steering the beam of an antenna array?

方程的应用(9.124) 指定增量长度 l 的大小:

$$l = \frac{n_0 u_p}{f_0} = \frac{9 \times 0.8 \times 3 \times 10^8}{10^{10}} = 21.6 \text{ cm}.$$

总之, 当 $N = 6$ 和 $kd = \pi$ 时, 方程: (9.121) 变为

$$F_{\text{an}}(\gamma') = \frac{\sin^2(3\gamma')}{36 \sin^2(\gamma'/2)}$$

和

$$\gamma' = kd(\cos \theta - \cos \theta_0) = \pi(\cos \theta - \cos \theta_0)$$

和

$$\cos \theta_0 = \frac{2n_0\pi}{kd} \left(\frac{\Delta f}{f_0} \right) = 18 \left(\frac{f - 10 \text{ GHz}}{10 \text{ GHz}} \right). \quad (9.130)$$

阵列方向图的形状与图9-30类似, 其主波束方向沿 $\theta = \theta_0$ 。对于 $f = f_0 = 10 \text{ GHz}$, $\theta_0 = 90^\circ$ (宽边方向); 对于 $f = 10.48 \text{ GHz}$, $\theta_0 = 30^\circ$; 对于 $f = 9.52 \text{ GHz}$, $\theta_0 = 150^\circ$ 。

对于 30° 和 150° 之间的任何其他 θ_0 值, 等式: (9.130) 提供了计算振荡器频率 f 所需值的方法。

Concept Question 9-11: Why are antenna arrays useful? Give examples of typical applications.

Concept Question 9-12: Explain how the pattern multiplication principle is used to compute the radiation pattern of an antenna array.

Concept Question 9-13: For a linear array, what roles do the array amplitudes and phases play?

Concept Question 9-14: Explain how electronic beam steering is accomplished.

Concept Question 9-15: Why is frequency scanning an attractive technique for steering the beam of an antenna array?

Chapter 9 Summary

Concepts

- An antenna is a transducer between a guided wave propagating on a transmission line and an EM wave propagating in an unbounded medium, or vice versa.
- Except for some solid-state antennas composed of non-linear semiconductors or ferrite materials, antennas are reciprocal devices; they exhibit the same radiation patterns for transmission as for reception.
- In the far-field region of an antenna, the radiated energy is approximately a plane wave.
- The electric field radiated by current antennas, such as wires, is equal to the sum of the electric fields radiated by all the Hertzian dipoles making up the antenna.
- The radiation resistance R_{rad} of a half-wave dipole is $73 \, \Omega$, which can be easily matched to a transmission line.
- The directional properties of an antenna are described by its radiation pattern, directivity, pattern solid angle, and half-power beamwidth.
- The Friis transmission formula relates the power received by an antenna to that transmitted by another antenna at a specified distance away.
- The far-zone electric field radiated by a large aperture (measured in wavelengths) is related to the field distribution across the aperture by Kirchhoff’s scalar diffraction theory. A uniform aperture distribution produces a far-field pattern with the narrowest possible beamwidth.
- By controlling the amplitudes and phases of the individual elements of an antenna array, it is possible to shape the antenna pattern and to steer the direction of the beam electronically.
- The pattern of an array of identical elements is equal to the product of the array factor and the antenna pattern of an individual antenna element.

第9章总结

概念

- 天线是传输线上传播的导波和无界介质中传播的电磁波之间的换能器, 反之亦然。
- 除一些由非线性半导体或铁氧体材料构成的固态天线外, 天线都是互易器件; 它们在传输和接收时表现出相同的辐射方向图。
- 在天线的远场区域, 辐射能量近似为平面波。
- 载流天线 (例如电线) 辐射的电场等于构成天线的所有赫兹偶极子辐射的电场之和。
- 半波偶极子的辐射电阻 R_{rad} 为 73Ω , 可以很容易地与传输线匹配。
- 天线的方向特性通过其辐射方向图、方向性、方向图立体角和半功率波束宽度来描述。
- 弗里斯传输公式将天线接收的功率与指定距离处的另一个天线发射的功率联系起来。
- 根据基尔霍夫标量衍射理论, 大孔径辐射的远区电场 (以波长测量) 与孔径上的场分布有关。均匀的孔径分布产生具有尽可能窄的波束宽度的远场方向图。
- 通过控制天线阵列中各个元件的幅度和相位, 可以塑造天线方向图并以电子方式控制波束方向。
- 相同元件的阵列的方向图等于阵列因子与单个天线元件的天线方向图的乘积。

Mathematical and Physical Models

Antenna Properties

Pattern solid angle	$\Omega_{\text{p}} = \iint_{4\pi} F(\theta, \phi) \, d\Omega$
Directivity	$D = \frac{4\pi}{\Omega_{\text{p}}}$
Gain	$G = \xi D, \quad \xi = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$
Effective area	$A_{\text{e}} = \frac{\lambda^2 D}{4\pi}$
Far-field distance	$R > \frac{2d^2}{\lambda}$

Short Dipole ($l \ll \lambda$)

$$\begin{aligned} \tilde{E}_{\theta} &= \frac{jI_0 l k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \\ \tilde{H}_{\phi} &= \frac{\tilde{E}_{\theta}}{\eta_0} \\ S(R, \theta) &= \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta \\ D &= 1.5 \\ \beta &= 90^\circ \\ R_{\text{rad}} &= 80\pi^2 (l/\lambda)^2 \end{aligned}$$

Friis Transmission Formula

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}} G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2 F_{\text{t}}(\theta_{\text{t}}, \phi_{\text{t}}) F_{\text{r}}(\theta_{\text{r}}, \phi_{\text{r}})$$

Antenna Arrays

Multiplication Principle

$$S(R_0, \theta, \phi) = S_{\text{e}}(R_0, \theta, \phi) F_{\text{a}}(\theta)$$

Uniform Phase

$$F_{\text{a}}(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2, \text{ with } \gamma = kd \cos \theta = \frac{2\pi d}{\lambda} \cos \theta$$

Linear Phase

$$F_{\text{a}}(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right|^2, \text{ with } \gamma' = kd \cos \theta - \delta$$

$\lambda/2$ Dipole

$$\begin{aligned} \tilde{E}_{\theta} &= j60I_0 \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\} \left(\frac{e^{-jkR}}{R} \right) \\ \tilde{H}_{\phi} &= \frac{\tilde{E}_{\theta}}{\eta_0} \\ S(R, \theta) &= \frac{15I_0^2}{\pi R^2} \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \\ D &= 1.64 \\ \beta &= 78^\circ \\ R_{\text{rad}} &\approx 73 \, \Omega \end{aligned}$$

Rectangular Aperture (Uniform)

$$\begin{aligned} S(R, \theta) &= S_0 \text{sinc}^2(\pi l_x \sin \theta / \lambda), \quad x\text{-}z \text{ plane} \\ S(R, \theta) &= S_0 \text{sinc}^2(\pi l_y \sin \theta / \lambda), \quad y\text{-}z \text{ plane} \\ \beta_{\text{xz}} &= 0.88 \frac{\lambda}{l_x}, \quad \beta_{\text{yz}} = 0.88 \frac{\lambda}{l_y} \\ D &= \frac{4\pi A_{\text{e}}}{\lambda^2} \approx \frac{4\pi A_{\text{p}}}{\lambda^2} \end{aligned}$$

数学与物理模型

天线参数

波束立体角	$\Omega_{\text{p}} = \iint_{4\pi} F(\theta, \phi) \, d\Omega$
方向性系数	$D = \frac{4\pi}{\Omega_{\text{p}}}$
增益	$G = \xi D, \quad \xi = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$
有效面积	$A_{\text{e}} = \frac{\lambda^2 D}{4\pi}$
远场距离	$R > \frac{2d^2}{\lambda}$

短电流元 ($l \ll \lambda$)

$$\begin{aligned} \tilde{E}_{\theta} &= \frac{jI_0 l k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \\ \tilde{H}_{\phi} &= \frac{\tilde{E}_{\theta}}{\eta_0} \\ S(R, \theta) &= \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \right) \sin^2 \theta \\ D &= 1.5 \\ \beta &= 90^\circ \\ R_{\text{rad}} &= 80\pi^2 (l/\lambda)^2 \end{aligned}$$

半波偶极子

$$\begin{aligned} \tilde{E}_{\theta} &= j60I_0 \left\{ \frac{\cos[(\pi/2) \cos \theta]}{\sin \theta} \right\} \left(\frac{e^{-jkR}}{R} \right) \\ \tilde{H}_{\phi} &= \frac{\tilde{E}_{\theta}}{\eta_0} \\ S(R, \theta) &= \frac{15I_0^2}{\pi R^2} \left\{ \frac{\cos^2[(\pi/2) \cos \theta]}{\sin^2 \theta} \right\} \\ D &= 1.64 \\ \beta &= 78^\circ \\ R_{\text{rad}} &\approx 73 \, \Omega \end{aligned}$$

弗里斯传输公式

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}} G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2 F_{\text{t}}(\theta_{\text{t}}, \phi_{\text{t}}) F_{\text{r}}(\theta_{\text{r}}, \phi_{\text{r}})$$

天线阵列

方向图相乘原理

$$S(R_0, \theta, \phi) = S_{\text{e}}(R_0, \theta, \phi) F_{\text{a}}(\theta)$$

同相激励

$$F_{\text{a}}(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2, \text{ 其中 } \gamma = kd \cos \theta = \frac{2\pi d}{\lambda} \cos \theta$$

线性相位递进

$$F_{\text{a}}(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma'} \right|^2, \text{ 其中 } \gamma' = kd \cos \theta - \delta$$

均匀照射的矩形口径

$$\begin{aligned} S(R, \theta) &= S_0 \text{sinc}^2(\pi l_x \sin \theta / \lambda), \quad x\text{-}z \text{ 平面} \\ S(R, \theta) &= S_0 \text{sinc}^2(\pi l_y \sin \theta / \lambda), \quad y\text{-}z \text{ 平面} \\ \beta_{\text{xz}} &= 0.88 \frac{\lambda}{l_x}, \quad \beta_{\text{yz}} = 0.88 \frac{\lambda}{l_y} \\ D &= \frac{4\pi A_{\text{e}}}{\lambda^2} \approx \frac{4\pi A_{\text{p}}}{\lambda^2} \end{aligned}$$

使用条件：短电流元一栏假设电流沿元长均匀。有效面积式 $A_{\text{e}}=\lambda^2\text{D}/(4\pi)$ 对应理想无损天线；一般天线应以增益 G 替代 D，并满足极化及阻抗匹配条件。

Important Terms	Provide definitions or explain the meaning of the following terms:	
3 dB beamwidth	elevation and azimuth planes	radiation efficiency ξ
antenna	elevation angle	radiation intensity
antenna array	end-fire direction	(normalized) $F(\theta, \phi)$
antenna directivity D	far-field (or far-zone) region	radiation lobes
antenna gain G	feeding coefficient	radiation pattern
antenna input impedance	frequency scanning	radiation resistance R_{rad}
antenna pattern	Friis transmission formula	reciprocal
antenna polarization	half-power beamwidth	scan angle
aperture distribution	isotropic antenna	short dipole (Hertzian dipole)
array distribution	linear phase distribution	signal-to-noise ratio S_{n}
array factor $F_{\text{a}}(\theta, \phi)$	loss resistance R_{loss}	solid angle
azimuth angle	null beamwidth	spherical propagation factor
beamwidth β	pattern multiplication principle	steradian
broadside direction	pattern solid angle Ω_{p}	system noise temperature T_{sys}
effective area	power density $\mathbf{S}(R, \theta, \phi)$	tapered aperture distribution
(effective aperture) A_{e}	Poynting vector	zenith angle
electronic steering	principal planes	

PROBLEMS

Sections 9-1 and 9-2: Short Dipole and Antenna Radiation Characteristics

***9.1** A center-fed Hertzian dipole is excited by a current $I_0 = 20$ A. If the dipole is $\lambda/50$ in length, determine the maximum radiated power density at a distance of 1 km.

9.2 A 50-cm long center-fed dipole directed along the z direction and located at the origin is excited by a 1 MHz source. If the current amplitude is $I_0 = 10$ A, determine:

- (a) the power density radiated at 2 km along the broadside of the antenna pattern, and
- (b) the fraction of the total power radiated within the sector between $\theta = 85^\circ$ and $\theta = 95^\circ$.

9.3 A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis?

- ***9.4** Determine:
- (a) the direction of maximum radiation,
 - (b) directivity,
 - (c) beam solid angle, and
 - (d) half-power beamwidth in the x - z plane,

* Answer(s) available in Appendix E.

Important Terms	Provide definitions or explain the meaning of the following terms:	
3 dB beamwidth	elevation and azimuth planes	radiation efficiency ξ
antenna	elevation angle	radiation intensity
antenna array	end-fire direction	(normalized) $F(\theta, \phi)$
antenna directivity D	far-field (or far-zone) region	radiation lobes
antenna gain G	feeding coefficient	radiation pattern
antenna input impedance	frequency scanning	radiation resistance R_{rad}
antenna pattern	Friis transmission formula	reciprocal
antenna polarization	half-power beamwidth	scan angle
aperture distribution	isotropic antenna	short dipole (Hertzian dipole)
array distribution	linear phase distribution	signal-to-noise ratio S_{n}
array factor $F_{\text{a}}(\theta, \phi)$	loss resistance R_{loss}	solid angle
azimuth angle	null beamwidth	spherical propagation factor
beamwidth β	pattern multiplication principle	steradian
broadside direction	pattern solid angle Ω_{p}	system noise temperature T_{sys}
effective area	power density $\mathbf{S}(R, \theta, \phi)$	tapered aperture distribution
(effective aperture) A_{e}	Poynting vector	zenith angle
electronic steering	principal planes	

问题

第 9-1 和 9-2 节：短偶极子和天线辐射特性

*9.1 中心馈电赫兹偶极子由电流 $I_0 = 20$ A 激励。如果偶极子长度为 $\lambda/50$ ，则确定 1 km 距离处的最大辐射功率密度。

9.2 沿 z 方向定向并位于原点的 50 厘米长的中心馈电偶极子由 1 MHz 源激励。如果电流幅度为 $I_0 = 10$ A，则确定：

(a) 沿天线方向图宽边 2 km 处辐射的功率密度，以及 (b) $\theta = 85^\circ$ 和 $\theta = 95^\circ$ 之间扇区内辐射总功率的比例。

9.3 一个 1 米长的偶极子被 1 MHz 电流激励，幅度为 12 A。该偶极子在距离偶极子轴 45° 方向 5 公里处辐射的平均功率密度是多少？

*9.4 确定：(a) 最大辐射方向，(b) 方向性，(c) 波束立体角，(d) x - z 平面内的半功率波束宽度，

9.5 对天线重复问题 9.4

9.6 2 米长的中心馈电偶极天线在 1 MHz 的 AM 广播频段工作。偶极子由半径为1毫米的铜线制成。

(a) 确定天线的辐射效率。

* (b) 天线增益（以分贝为单位）是多少？(c) 需要多大的天线电流才能使天线辐射 80 W，发电机必须向天线提供多少功率？

9.7 对于工作频率为 5 MHz 的 20 厘米长天线，重复问题 9.6。

9.8 确定短偶极子辐射效率的频率依赖性，并在 600 kHz 至 60 MHz 范围内绘制它。偶极子由铜制成，长度为10厘米，圆形横截面的半径为1毫米。

* Answer(s) available in Appendix E.

对于其归一化辐射强度由下式给出的天线

$$F(\theta, \phi) = \begin{cases} 1 & \text{for } 0 \leq \theta \leq 60^\circ \text{ and } 0 \leq \phi \leq 2\pi \\ 0 & \text{elsewhere.} \end{cases}$$

建议：在计算所需数量之前绘制图案草图。

$$F(\theta, \phi) = \begin{cases} \sin^2 \theta \cos^2 \phi & \text{for } 0 \leq \theta \leq \pi \\ & \text{and } -\pi/2 \leq \phi \leq \pi/2 \\ 0 & \text{elsewhere.} \end{cases}$$

9.6 2 米长的中心馈电偶极天线在 1 MHz 的 AM 广播频段工作。偶极子由半径为1毫米的铜线制成。

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***9.9** An antenna with a pattern solid angle of 1.5 (sr) radiates 60 W of power. At a range of 1 km, what is the maximum power density radiated by the antenna?

9.10 An antenna with a radiation efficiency of 90% has a directivity of 7.0 dB. What is its gain in decibels?

***9.11** The radiation pattern of a circular parabolic-reflector antenna consists of a circular major lobe with a half-power beamwidth of 3° and a few minor lobes. Ignoring the minor lobes, obtain an estimate for the antenna directivity in dB.

9.12 The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \quad \text{for } 0 \leq \theta \leq \pi$$

where θ is in radians. Determine:

- (a) the half-power beamwidth,
- (b) the pattern solid angle, and
- (c) the antenna directivity.

Sections 9-3 and 9-4: Dipole Antennas

9.13 Repeat Problem 9.6 for a 1-m long half-wave dipole that operates in the FM/TV broadcast band at 150 MHz.

***9.14** Assuming the loss resistance of a half-wave dipole antenna to be negligibly small and ignoring the reactance component of its antenna impedance, calculate the standing-wave ratio on a $50\ \Omega$ transmission line connected to the dipole antenna.

9.15 A 50-cm long dipole is excited by a sinusoidally varying current with an amplitude $I_0 = 5$ A. Determine the total radiated power if the oscillating frequency is:

- (a) 1 MHz,
- (b) 300 MHz.

9.16 For a short dipole with length l where $l \ll \lambda$, instead of treating the current $\tilde{I}(z)$ as constant along the dipole, as was done in Section 9-1, a more realistic approximation that ensures the current goes to zero at the dipole ends is to describe $\tilde{I}(z)$ by the triangular function

$$\tilde{I}(z) = \begin{cases} I_0(1 - 2z/l) & \text{for } 0 \leq z \leq l/2 \\ I_0(1 + 2z/l) & \text{for } -l/2 \leq z \leq 0 \end{cases}$$

shown in **Fig. P9.16**. Use this current distribution to determine the following:

- *9.16** (a) the far-field $\tilde{\mathbf{E}}(R, \theta, \phi)$,
- (b) the power density $S(R, \theta, \phi)$,

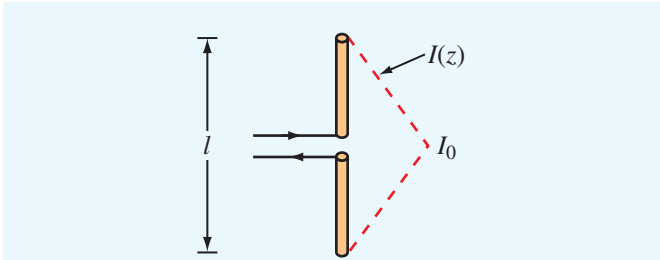


Figure P9.16 Triangular current distribution on a short dipole (Problem 9.16).

- (c) the directivity D , and
- (d) the radiation resistance R_{rad} .

9.17 For a dipole antenna of length $l = 3\lambda/2$,

- *9.17** (a) determine the directions of maximum radiation,
- (b) obtain an expression for S_{max} ,
- (c) generate a plot of the normalized radiation pattern $F(\theta)$, and
- (d) compare your pattern with that shown in **Fig. 9-17(c)**.

9.18 For a dipole antenna of length $l = \lambda/4$,

- (a) determine the directions of maximum radiation,
- (b) obtain an expression for S_{max} , and
- (c) generate a plot of the normalized radiation pattern $F(\theta)$.

9.19 Repeat parts (a)–(c) of Problem 9.17 for a dipole of length $l = 3\lambda/4$.

***9.20** Repeat parts (a)–(c) of Problem 9.17 for a dipole of length $l = \lambda$.

9.21 A car antenna is a vertical monopole over a conducting surface. Repeat Problem 9.6 for a 1-m long car antenna operating at 1 MHz. The antenna wire is made of aluminum with $\mu_c = \mu_0$ and $\sigma_c = 3.5 \times 10^7$ S/m, and its diameter is 1 cm.

9.22 For a dipole antenna of length $l = 2.5\lambda$, use Module 9.2 to determine:

- (a) approximate directions of maximum radiation,
- (b) antenna directivity, and
- (c) antenna radiation resistance.

9.23 For a dipole antenna of length $l = 3\lambda$, use Module 9.2 to determine:

- (a) approximate directions of maximum radiation,
- (b) antenna directivity, and
- (c) antenna radiation resistance.

***9.9** 方向图立体角为 1.5 (sr) 的天线辐射 60 W 功率。在 1 公里范围内，天线辐射的最大功率密度是多少？

9.10 辐射效率为 90% 的天线的方向性为 7.0 dB。它的增益是多少分贝？

***9.11** 圆形抛物面反射器天线的辐射方向图由半功率波束宽度为 3° 的圆形主瓣和几个小瓣组成。忽略小瓣，获得天线方向性的估计值（以 dB 为单位）。

9.12 某个天线的归一化辐射强度由下式给出

$$F(\theta) = \exp(-20\theta^2) \quad \text{for } 0 \leq \theta \leq \pi$$

其中 θ 的单位是弧度。决定：

- (a) 半功率波束宽度，
- (b) 图案立体角，以及
- (c) 天线方向性。

第 9-3 节和第 9-4 节：偶极天线

9.13 对于工作在 150 MHz FM/TV 广播频段的 1 米长半波偶极子，重复问题 9.6。

***9.14** 假设半波偶极子天线的损耗电阻小到可以忽略不计，并忽略其天线阻抗的电抗分量，计算连接到偶极子天线的 $50\ \Omega$ 传输线上的驻波比。

9.15 A 50 厘米长的偶极子由振幅为 $I_0 = 5$ A 的正弦变化电流激励。如果振荡频率为：，则确定总辐射功率：

- (a) 1 MHz, (
- b) 300 MHz

9.16 对于长度为 l 的短偶极子，其中 $l \ll \lambda$ ，不是将电流 $\tilde{I}(z)$ 视为沿偶极子的常数，如第 9-1 节中所做的那样，确保电流在偶极子末端变为零的更现实的近似是通过三角函数来描述 $\tilde{I}(z)$

$$\tilde{I}(z) = \begin{cases} I_0(1 - 2z/l) & \text{for } 0 \leq z \leq l/2 \\ I_0(1 + 2z/l) & \text{for } -l/2 \leq z \leq 0 \end{cases}$$

如图P9.16所示。使用此电流分布可以确定以下内容：

- *9.16** (a) 远场 $\tilde{\mathbf{E}}(R, \theta, \phi)$,
- (b) 功率密度 $S(R, \theta, \phi)$,

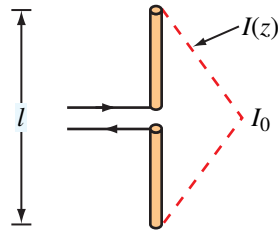


图 P9.16 短偶极子上的三角形电流分布（问题 9.16）。

- (c)方向性D，以及
- (d)辐射电阻 R_{rad} 。

9.17 对于长度为 $l = 3\lambda/2$ 的偶极天线，

- *9.17** (a) 确定最大辐射方向，
- (b) 获得 S_{max} 的表达式，(c) 生成归一化辐射方向图 $F(\theta)$ 的图，以及 (d) 将您的方向图与图 9-17(c) 中所示的方向图进行比较。

9.18 对于长度为 $l = \lambda/4$ 的偶极天线，

- (a) 确定最大辐射方向，
- (b) 获得 S_{max} 的表达式，并且
- (c) 生成归一化辐射方向图 $F(\theta)$ 。

9.19 对于长度为 $l = 3\lambda/4$ 的偶极子，重复问题 9.17 的 (a)–(c) 部分。

***9.20** 对于长度为 $l = \lambda$ 的偶极子，重复问题 9.17 的 (a)–(c) 部分。

9.21 汽车天线是导电表面上的垂直单极天线。对于工作频率为 1 MHz 的 1 米长汽车天线，重复问题 9.6。天线线材质为 $\mu_c = \mu_0$ 和 $\sigma_c = 3.5 \times 10^7$ S/m的铝，直径为1 cm。

9.22 对于长度为 $l = 2.5\lambda$ 的偶极天线，使用模块 9.2 确定：

- (a) 最大辐射的大致方向，
- (b) 天线方向性，以及 (c) 天线辐射电阻。

9.23 对于长度为 $l = 3\lambda$ 的偶极天线，使用模块 9.2 确定：

- (a) 最大辐射的大致方向，
- (b) 天线方向性，以及
- (c) 天线辐射电阻。

9.24 A dipole antenna has a radiation impedance

$$Z_{\text{rad}} = R_{\text{rad}} + jX_{\text{rad}}.$$

For a dipole of length $l = \lambda/2$, $R_{\text{rad}} = 73.13 \, \Omega$ and $X_{\text{rad}} = 42.55 \, \Omega$. Use Module 9.3 to modify the value of l slightly so as to reduce X_{rad} to less than $1 \, \Omega$.

- (a) What is the value of l ?
 (b) What is the corresponding value of R_{rad} ?

Sections 9-5 and 9-6: Effective Area and Friis Formula

9.25 Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm.

*9.26 A 3 GHz line-of-sight microwave communication link consists of two lossless parabolic dish antennas, each 1 m in diameter. If the receive antenna requires 10 nW of receive power for good reception and the distance between the antennas is 40 km, how much power should be transmitted?

9.27 A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km?

*9.28 A 150 MHz communication link consists of two vertical half-wave dipole antennas separated by 2 km. The antennas are lossless, the signal occupies a bandwidth of 3 MHz, the system noise temperature of the receiver is 600 K, and the desired signal-to-noise ratio is 17 dB. What transmitter power is required?

9.29 Consider the communication system shown in Fig. P9.29 with all components properly matched. Determine the following if $P_t = 10 \, \text{W}$ and $f = 6 \, \text{GHz}$.

- (a) What is the power density at the receiving antenna (assuming proper alignment of antennas)?
 (b) What is the received power?
 (c) If $T_{\text{sys}} = 1,000 \, \text{K}$ and the receiver bandwidth is 20 MHz, what is the signal-to-noise ratio in decibels?

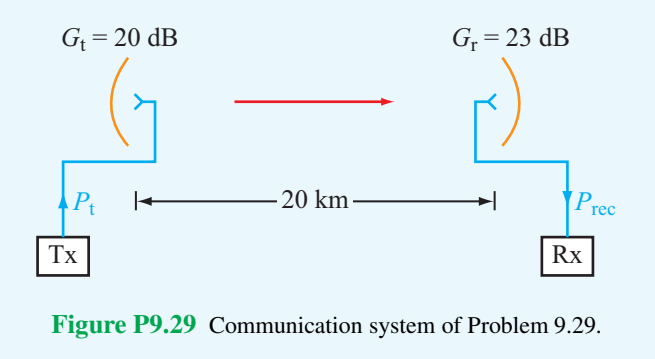


Figure P9.29 Communication system of Problem 9.29.

9.30 The configuration shown in Fig. P9.30 depicts two vertically oriented half-wave dipole antennas pointed towards each other with both positioned on 100-m tall towers separated by a distance of 5 km. If the transmit antenna is driven by a 50 MHz current with amplitude $I_0 = 2 \, \text{A}$, determine the following:

- (a) The power received by the receive antenna in the absence of the surface. (Assume both antennas to be lossless.)
 (b) The power received by the receive antenna after incorporating reflection by the ground surface, assuming the surface to be flat and to have $\epsilon_r = 9$ and conductivity $\sigma = 10^{-3} \, (\text{S/m})$.

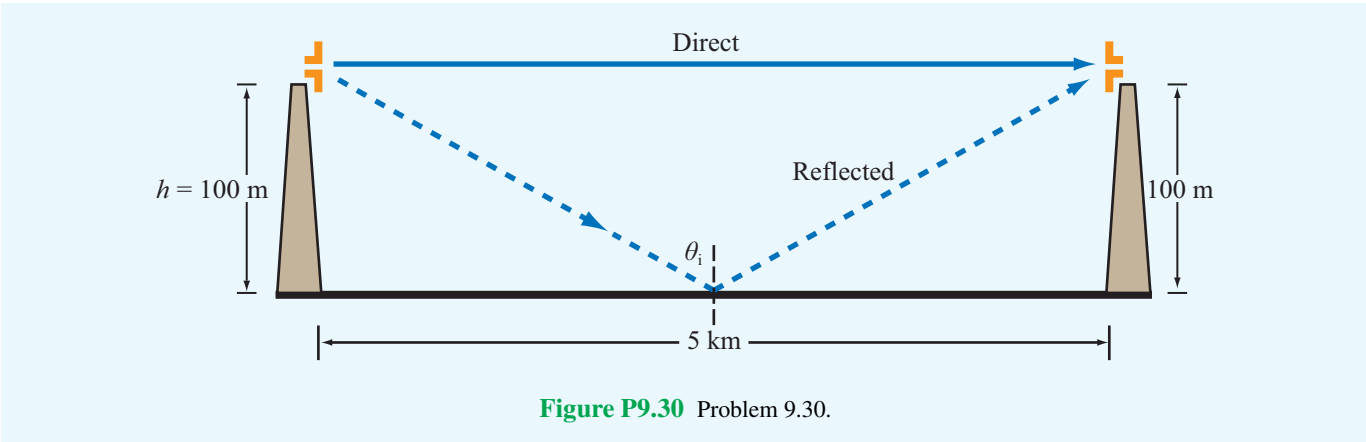


Figure P9.30 Problem 9.30.

9.24 偶极天线具有辐射阻抗

$$Z_{\text{rad}} = R_{\text{rad}} + jX_{\text{rad}}.$$

对于长度为 $l = \lambda/2$ 的偶极子, $R_{\text{rad}} = 73.13 \, \Omega$ 和 $X_{\text{rad}} = 42.55 \, \Omega$ 。使用模块9.3稍微修改 l 的值, 使 X_{rad} 减少到小于 $1 \, \Omega$ 。

- (a) l 的值是多少? (b) R_{rad} 对应的值是多少?

第 9-5 节和第 9-6 节: 有效面积和弗里斯公式

9.25 确定半波偶极子天线在 100 MHz 下的有效面积, 并与线径为 2 cm 时的物理横截面进行比较。

*9.26 3 GHz 视距微波通信链路由两个无损抛物面碟形天线组成, 每个直径为 1 m。如果接收天线需要 10 nW 的接收功率才能获得良好的接收效果, 并且天线之间的距离为 40 km, 则应发射多少功率?

9.27 半波偶极子电视广播天线在 50 MHz 下发射 1 kW 功率。如果位于 30 公里的距离, 增益为 3 dB 的家用电视天线接收到的功率是多少?

*9.28 150 MHz 通信链路由两个相距 2 km 的垂直半波偶极天线组成。天线无损, 信号占用带宽 3 MHz, 接收机系统噪声温度 600 K, 期望信噪比 17 dB。需要多大的发射机功率?

9.29 考虑图 P9.29 所示的通信系统, 所有组件均正确匹配。如果 $P_t = 10 \, \text{W}$ 和 $f = 6 \, \text{GHz}$, 请确定以下内容。

- (a) 接收天线的功率密度是多少 (假设天线正确对准)?
 (b) 接收功率是多少? (c) 如果 $T_{\text{sys}} = 1,000 \, \text{K}$ 且接收器带宽为 20 MHz, 则信噪比 (以分贝为单位) 是多少?

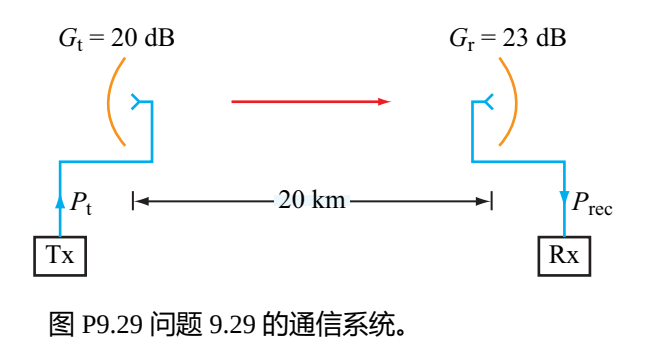


图 P9.29 问题 9.29 的通信系统。

9.30 图 P9.30 所示的配置描绘了两个垂直定向的半波偶极子天线, 它们彼此指向, 且均位于 100 米高的塔上, 塔上相距 5 公里。如果中转天线由幅度为 $I_0 = 2 \, \text{A}$ 的 50 MHz 电流驱动, 请确定以下内容:

- (a) 在没有表面的情况下接收天线接收到的功率。(假设两个天线都是无损的。)
 (b) 接收天线在纳入地面反射后接收到的功率, 假设地面平坦且具有 $\epsilon_r = 9$ 和电导率 $\sigma = 10^{-3} \, (\text{S/m})$ 。

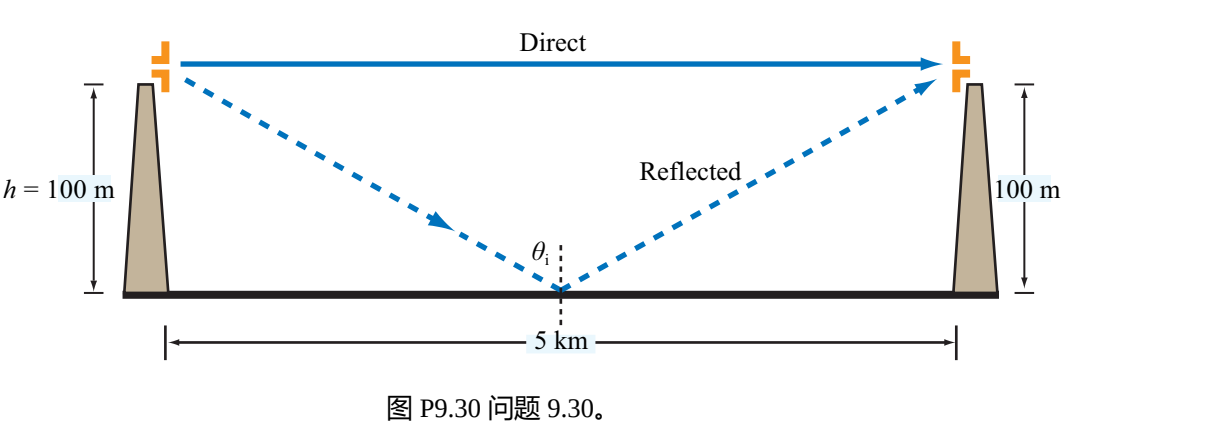


图 P9.30 问题 9.30。

9.31 Fig. P9.31 depicts a half-wave dipole connected to a generator through a matched transmission line. The directivity of the dipole can be modified by placing a reflecting rod a distance d behind the dipole. What would its reflectivity in the forward direction be if

(a) $d = \lambda/4$
(b) $d = \lambda/2$

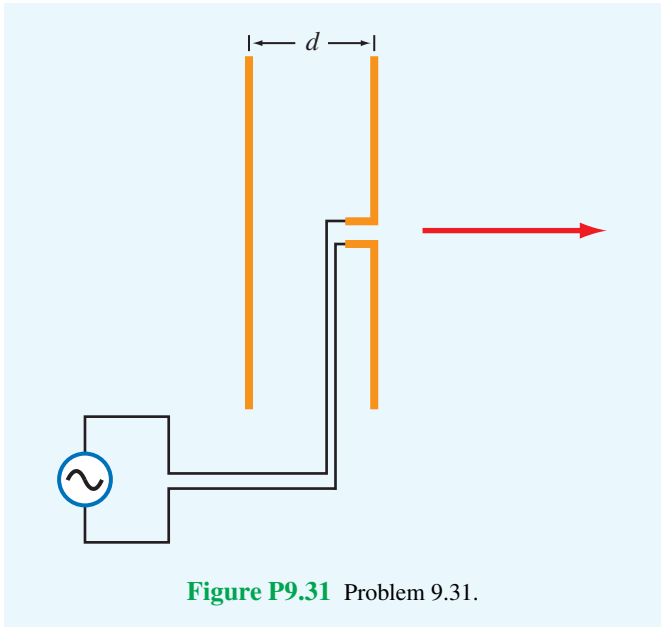


Figure P9.31 Problem 9.31.

9.32 The configuration shown in Fig. P9.32 depicts a satellite repeater with two antennas, one pointed towards the antenna of ground station 1 and the other towards the antenna of ground station 2. All antennas are parabolic dishes; antennas A_1 and A_4 are each 4 m in diameter, antennas A_2 and A_3 are

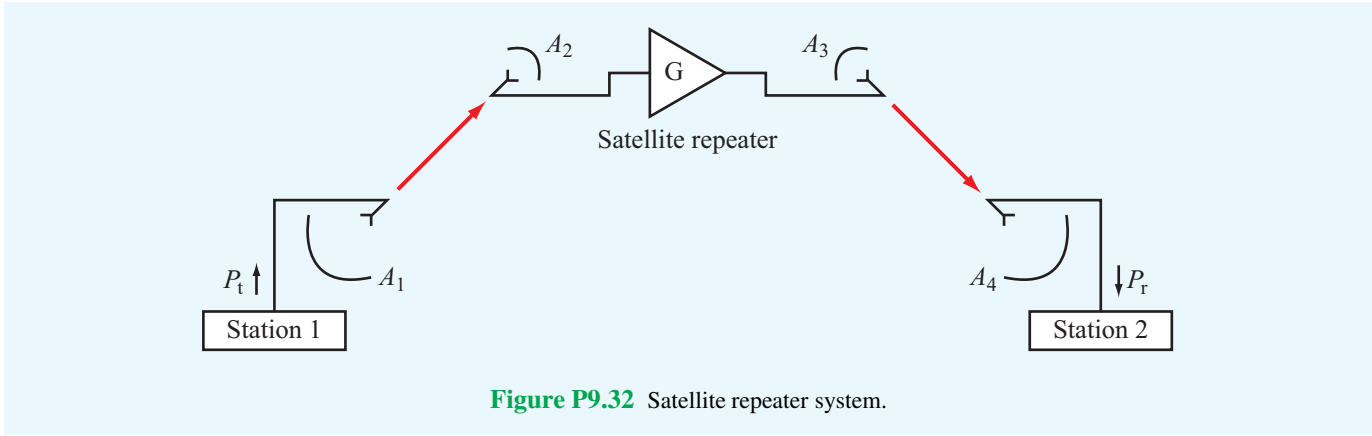


Figure P9.32 Satellite repeater system.

each 2 m in diameter, and the distance between the satellite and each of the ground stations is 40,000 km. Upon receiving the signal by its antenna A_2 , the satellite transponder boosts the power gain by 80 dB and then retransmits the signal to A_4 . The system operates at 10 GHz with $P_t = 1$ kW. Determine the received power P_r . Assume all antennas to be lossless.

Sections 9-7 and 9-8: Radiation by Apertures

- *9.33** A uniformly illuminated aperture is of length $l_x = 20\lambda$. Determine the beamwidth between first nulls in the x - z plane.
- 9.34** The 10 dB beamwidth is the beam size between the angles at which $F(\theta)$ is 10 dB below its peak value. Determine the 10 dB beamwidth in the x - z plane for a uniformly illuminated aperture with length $l_x = 10\lambda$.
- *9.35** A uniformly illuminated rectangular aperture situated in the x - y plane is 2 m high (along x) and 1 m wide (along y). If $f = 10$ GHz, determine:
- (a) the beamwidths of the radiation pattern in the elevation plane (x - z plane) and the azimuth plane (y - z plane), and
(b) the antenna directivity D in decibels.
- 9.36** An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz.
- (a) What is the antenna directivity in dB?
(b) If the antenna area is doubled, what will be the new directivity and new beamwidth?
(c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then?

9.31 图 P9.31 描绘了通过匹配传输线连接到发电机的半波偶极子。偶极子的方向性可以通过在偶极子后面距离 d 处放置一根反射棒来改变。如果它向前的反射率是多少

(a) $d = \lambda/4$
(b) $d = \lambda/2$

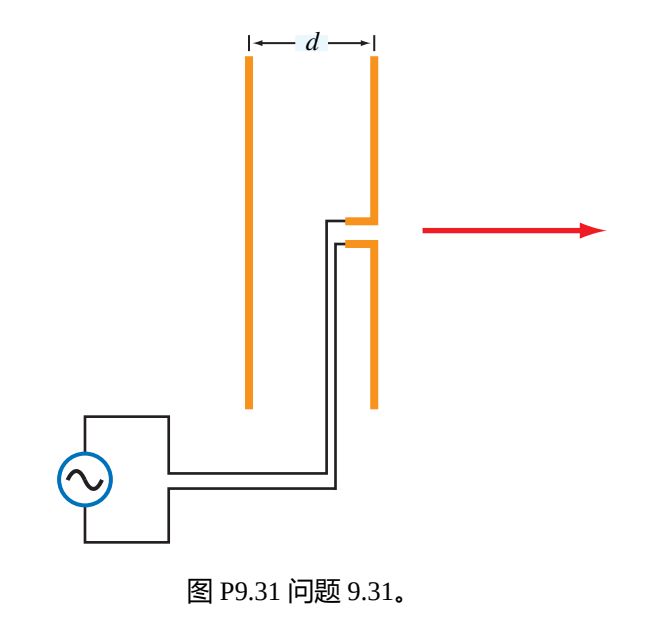


图 P9.31 问题 9.31。

9.32 图 P9.32 所示的配置描述了一个带有两根天线的卫星中继器，一根指向地面站 1 的天线，另一根指向地面站 2 的天线。所有天线都是抛物面天线；天线 A_1 和 A_4 的直径均为 4 m，天线 A_2 和 A_3 的直径为

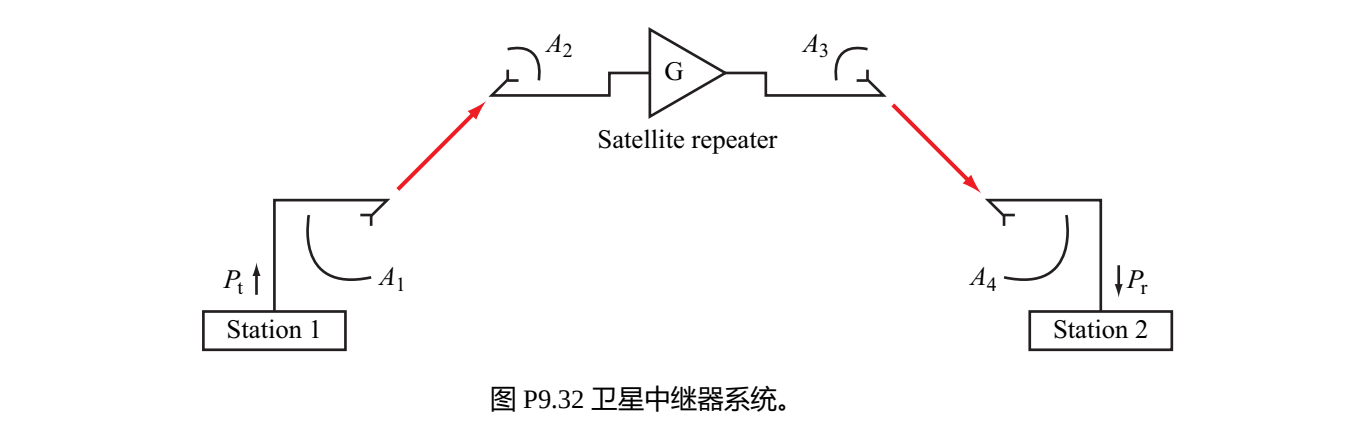


图 P9.32 卫星中继器系统。

每个卫星直径为 2 m，卫星与每个地面站的距离为 4 万公里。卫星转发器通过天线 A_2 接收到信号后，将功率增益提高 80 dB，然后将信号重新发送到 A_4 。该系统工作频率为 10 GHz， $P_t = 1$ kW。确定接收功率 P_r 。假设所有天线都是无损的。

第 9-7 节和 9-8 节：孔径辐射

- *9.33** 均匀照明孔径的长度为 $l_x = 20\lambda$ 。确定 x - z 平面中第一个零点之间的波束宽度。
- 9.34** 10 dB 波束宽度是 $F(\theta)$ 低于其峰值 10 dB 的角度之间的波束大小。确定长度为 $l_x = 10\lambda$ 的均匀照明孔径在 x - z 平面上的 10 dB 波束宽度。
- *9.35** 位于 x - y 平面上的均匀照明矩形孔径为 2 m 高（沿 x ）和 1 m 宽（沿 y ）。如果 $f = 10$ GHz，则确定：
- (a) 仰角平面（ x - z 平面）和方位角平面（ y - z 平面）中辐射方向图的波束宽度，以及 (b) 天线方向性 D （以分贝为单位）。
- 9.36** 圆形孔径天线的圆形波束在 20 GHz 时波束宽度为 3° 。
- (a) 天线方向性（以 dB 为单位）是多少？(b) 如果天线面积加倍，新的方向性和新的波束宽度将是多少？(c) 如果孔径保持与 (a) 相同，但频率加倍至 40 GHz，那么方向性和波束宽度会变成什么？

9.37 Compare directivity D_{ant} of a 1-m diameter antenna aperture operating at 10 GHz with directivity D_{eye} of the eye’s pupil operating in the middle of the visible spectrum at $\lambda = 0.5\text{ }\mu\text{m}$. Treat the pupil as a circular aperture with a diameter of 4 mm.

***9.38** A 94 GHz automobile collision-avoidance radar uses a rectangular-aperture antenna placed above the car’s bumper. If the antenna is 1 m in length and 10 cm in height, determine:

- (a) its elevation and azimuth beamwidths, and
- (b) the horizontal extent of the beam at a distance of 300 m.

9.39 A microwave telescope consisting of a very sensitive receiver connected to a 100 m parabolic-dish antenna is used to measure the energy radiated by astronomical objects at 20 GHz. If the antenna beam is directed toward the moon and the moon extends over a planar angle of 0.5° from Earth, what fraction of the moon’s cross-section will be occupied by the beam?

9.40 A parabolic dish antenna has a diameter $d = 5\lambda$. Use Module 9.4 to evaluate three antenna parameters, namely the directivity D , the beamwidth β , and the level of the first sidelobe next to the mainbeam, all at the following three values of the taper factor: $\alpha = 0$, $\alpha = 0.5$, and $\alpha = 1$.

9.41 Given a parabolic dish antenna with a diameter $d = 10\lambda$, use Module 9.4 to select the highest value of the taper factor α without having the beamwidth exceed 6° .

Sections 9-9 and 9-11: Antenna Arrays

9.42 A two-element array consisting of two isotropic antennas separated by a distance d along the z axis is placed in a coordinate system whose z axis points eastward and whose x axis points toward the zenith. If a_0 and a_1 are the amplitudes of the excitations of the antennas at $z = 0$ and at $z = d$, respectively, and if δ is the phase of the excitation of the antenna at $z = d$ relative to that of the other antenna, find the array factor and plot the pattern in the x – z plane for the following.

- * (a)** $a_0 = a_1 = 1$, $\delta = \pi/4$, and $d = \lambda/2$
- (b)** $a_0 = 1$, $a_1 = 2$, $\delta = 0$, and $d = \lambda$
- (c)** $a_0 = a_1 = 1$, $\delta = -\pi/2$, and $d = \lambda/2$
- (d)** $a_0 = 1$, $a_1 = 2$, $\delta = \pi/4$, and $d = \lambda/2$
- (e)** $a_0 = 1$, $a_1 = 2$, $\delta = \pi/2$, and $d = \lambda/4$

9.43 If the antennas in part (a) of Problem 9.42 are parallel, vertical, Hertzian dipoles with axes along the x direction, determine the normalized radiation intensity in the x – z plane and plot it.

***9.44** Consider the two-element dipole array of **Fig. 9-29(a)**. If the two dipoles are excited with identical feeding coefficients ($a_0 = a_1 = 1$ and $\psi_0 = \psi_1 = 0$), choose (d/λ) so that the array factor has a maximum at $\theta = 45^\circ$.

9.45 Choose (d/λ) so that the array pattern of the array of Problem 9.44 has a null, rather than a maximum, at $\theta = 45^\circ$.

9.46 Find and plot the normalized array factor and determine the half-power beamwidth for a five-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is $3\lambda/4$.

9.47 Repeat Problem 9.46, but change the excitation to tapered amplitude distribution so that the amplitude of the central element is 1, the amplitudes of the next adjacent elements are both 0.5, and those of the outer elements are both 0.25.

9.48 Repeat Problem 9.46 for a nine-element array.

***9.49** A five-element equally spaced linear array with $d = \lambda/2$ is excited with uniform phase and an amplitude distribution given by the binomial distribution

$$a_i = \frac{(N-1)!}{i!(N-i-1)!}, \quad i = 0, 1, \dots, (N-1),$$

where N is the number of elements. Develop an expression for the array factor.

9.50 A three-element linear array of isotropic sources aligned along the z axis has an interelement spacing of $\lambda/4$ (**Fig. P9.50**). The amplitude excitation of the center element is twice that of the bottom and top elements, and the phases are $-\pi/2$ for the bottom element and $\pi/2$ for the top element relative to that of the center element. Determine the array factor and plot it in the elevation plane.

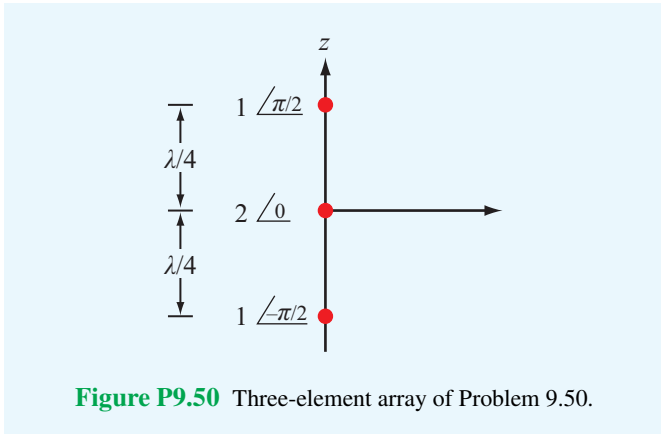


Figure P9.50 Three-element array of Problem 9.50.

9.37 比较在 10 GHz 工作的 1 米直径天线孔径的方向性 D_{ant} 与在可见光谱中间工作在 $\lambda = 0.5\text{ }\mu\text{m}$ 的眼睛瞳孔的方向性 D_{eye} 。将瞳孔视为直径为 4 mm 的圆形光圈。

***9.38** 94 GHz 汽车防撞雷达使用放置在汽车保险杠上方的矩形孔径天线。如果天线长 1 m，高 10 cm，则求：

- (a) 其仰角和方位角波束宽度，以及 (b) 300 m 距离处波束的水平范围。

9.39 微波望远镜由连接到 100 m 抛物面天线的非常灵敏的接收器组成，用于测量天体在 20 GHz 辐射的能量。如果天线波束指向月球，并且月球与地球的平面角为 0.5° ，则波束将占据月球横截面的几分之一？

9.40 抛物面碟形天线的直径为 $d = 5\lambda$ 。使用模块 9.4 评估三个天线参数，即方向性 D 、波束宽度 β 和主波束旁边的第一个旁瓣的电平，所有参数均采用以下三个锥度因子值： $\alpha = 0$ 、 $\alpha = 0.5$ 和 $\alpha = 1$ 。

9.41 给定直径为 $d = 10\lambda$ 的抛物面天线，使用模块 9.4 选择锥度系数 α 的最高值，且波束宽度不超过 6° 。

第 9-9 和 9-11 节：天线阵列

9.42 由沿 z 轴间隔距离 d 的两个各向同性天线组成的二元阵列放置在 z 轴指向东且 x 轴指向天顶的坐标系中。如果 a_0 和 a_1 分别是天线在 $z = 0$ 和 $z = d$ 处的激励幅度，并且如果 δ 是天线在 $z = d$ 处相对于另一个天线的激励相位，则求出阵列因子并在 x - z 平面上绘制以下方向图。

- * (a)** $a_0 = a_1 = 1$ 、 $\delta = \pi/4$ 和 $d = \lambda/2$ (b) $a_0 = 1$ 、 $a_1 = 2$ 、 $\delta = 0$ 和 $d = \lambda$ (c) $a_0 = a_1 = 1$ 、 $\delta = -\pi/2$ 和 $d = \lambda/2$ (d) $a_0 = 1$ 、 $a_1 = 2$ 、 $\delta = \pi/4$ 和 $d = \lambda/2$ (e) $a_0 = 1$ 、 $a_1 = 2$ 、 $\delta = \pi/2$ 和 $d = \lambda/4$

9.43 如果问题 9.42 (a) 部分中的天线是平行、垂直、轴沿 x 方向的赫兹偶极子，则确定 x - z 平面中的归一化辐射强度并将其绘制出来。

***9.44** 考虑图 9-29(a) 的二元偶极子阵列。如果两个偶极子以相同的馈电系数 ($a_0 = a_1 = 1$ 和 $\psi_0 = \psi_1 = 0$) 激发，请选择 (d/λ) ，以便阵列因子在 $\theta = 45^\circ$ 处具有最大值。

9.45 选择 (d/λ) ，使得问题 9.44 的数组的数组模式在 $\theta = 45^\circ$ 处具有空值，而不是最大值。

9.46 求并绘制归一化阵列因子，并确定以等相位和均匀幅度分布激励的五元线性阵列的半功率波束宽度。元素间间距为 $3\lambda/4$ 。

9.47 重复问题 9.46，但将激励改为锥形振幅分布，使得中心单元的振幅为 1，接下来的相邻单元的振幅均为 0.5，外部单元的振幅均为 0.25。

9.48 对九元素数组重复问题 9.46。

***9.49** 具有 $d = \lambda/2$ 的五元等距线性阵列由均匀相位和由二项式分布给出的幅度分布激励

$$a_i = \frac{(N-1)!}{i!(N-i-1)!}, \quad i = 0, 1, \dots, (N-1),$$

其中 N 是元素的数量。开发数组因子的表达式。

9.50 沿 z 轴排列的各向同性源的三元线性阵列的元间间距为 $\lambda/4$ (图 P9.50)。中心单元的振幅激励是底部和顶部单元的两倍，相对于中心单元的相位，底部单元的相位为 $-\pi/2$ ，顶部单元的相位为 $\pi/2$ 。确定阵列系数并将其绘制在高程平面上。

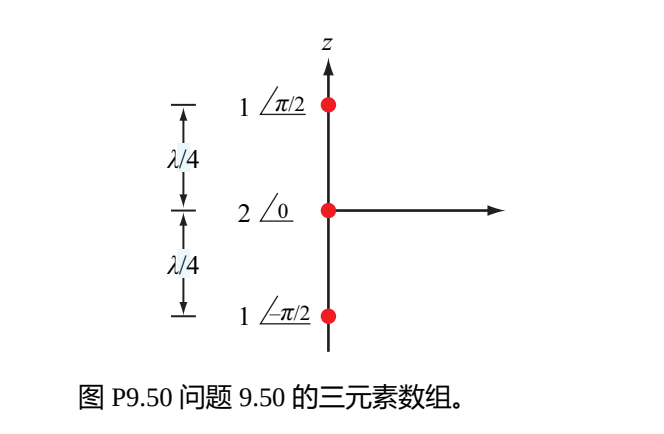


图 P9.50 问题 9.50 的三元素数组。

***9.51** An eight-element linear array with $\lambda/2$ spacing is excited with equal amplitudes. To steer the main beam to a direction 60° below the broadside direction, what should be the incremental phase delay between adjacent elements? Also, give the expression for the array factor and plot the pattern.

9.52 A linear array arranged along the z axis consists of 12 equally spaced elements with $d = \lambda/2$. Choose an appropriate incremental phase delay δ so as to steer the main beam to a direction 30° above the broadside direction. Provide an

expression for the array factor of the steered antenna and plot the pattern. From the pattern, estimate the beamwidth.

9.53 Use Module 9.5 to obtain the radiation pattern for a two-dipole array, each 0.5λ in length, separated by $\lambda/2$, and fed equally in amplitude but with one of them 90° ahead of the other in phase.

9.54 Repeat Problem 9.53, but change the separation between the two dipoles to $d = 1\lambda$.

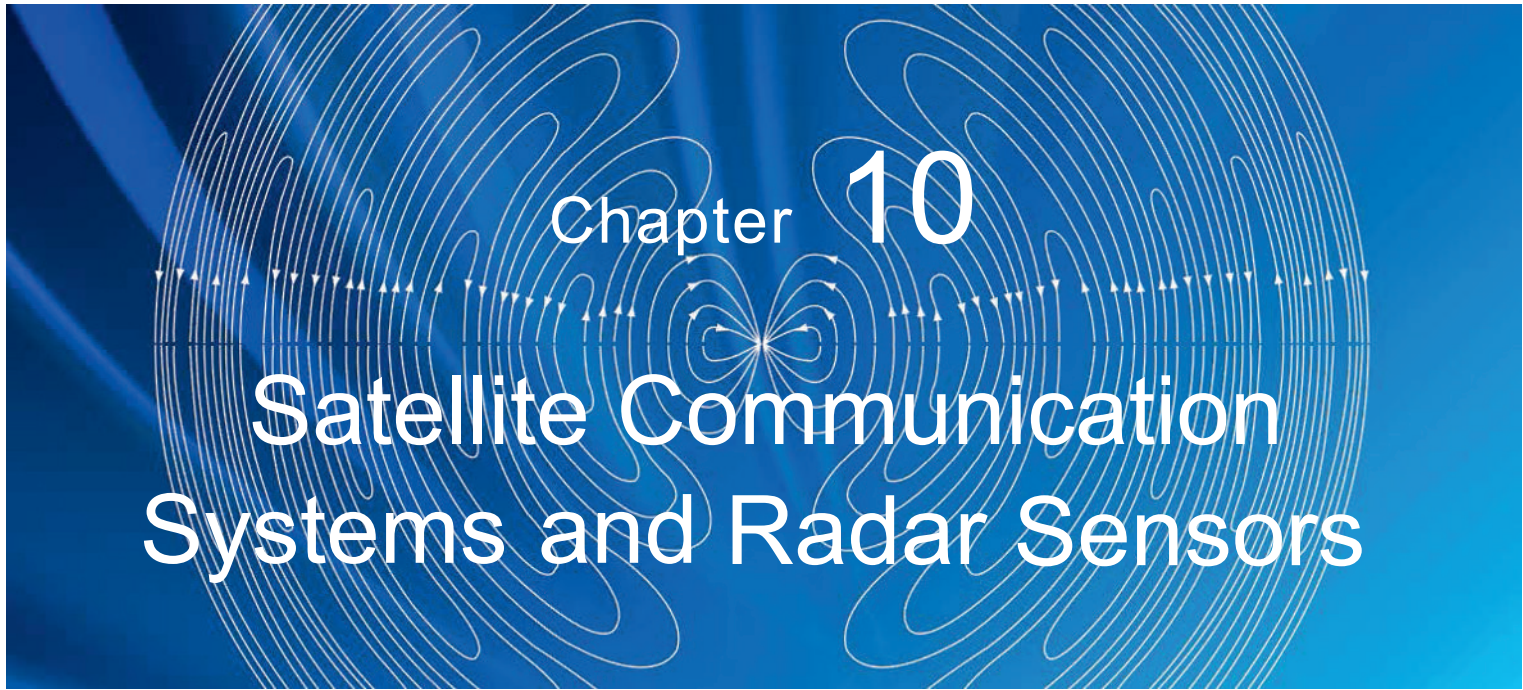
***9.51** 具有 $\lambda/2$ 间距的八元线性阵列以等幅激励。为了将主波束转向侧射方向下方 60° 的方向，相邻元件之间的增量相位延迟应该是多少？另外，给出数组因子的表达式并绘制模式。

9.52 沿 z 轴排列的线性阵列由 12 个等间隔的元素组成，元素的间距为 $d = \lambda/2$ 。选择适当的增量相位延迟 δ ，以便将主波束转向高于侧射方向 30° 的方向。提供一个

可控天线阵列因子的表达式并绘制方向图。根据该图案，估计波束宽度。

9.53 使用模块 9.5 获得双偶极子阵列的辐射方向图，每个偶极子阵列长度为 0.5λ ，间隔为 $\lambda/2$ ，并且馈电幅度相等，但其中一个在相位上领先另一个 90° 。

9.54 重复问题 9.53， 但将两个偶极子之间的间隔更改为 $d = 1\lambda$ 。



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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Describe the basic operation of satellite transponders.
2. Calculate the power budget for a communication link.
3. Describe how radar attains spatial and angular resolutions, calculate the maximum detectable range, and explain the trade-off between the probabilities of detection and false alarm.
4. Calculate the Doppler frequency shift observed by a radar.
5. Describe the monopulse-radar technique.



章节内容

应用示例、435 10-1 卫星通信系统、435 10-2 卫星转发器、436 10-3 通信链路功率预算、439 10-4 天线波束、440 10-5 雷达传感器、441 10-6 目标检测、443 10-7 多普勒雷达、445 10-8 单脉冲雷达、447 第 10 章总结、449 问题、450

目标

学习本章介绍的材料后，您应该能够：

1. 描述卫星转发器的基本操作。
2. 计算通信链路的功率预算。
3. 描述雷达如何获得空间和角度分辨率，计算最大可检测范围，并解释检测概率和误报概率之间的权衡。
4. 计算雷达观测到的多普勒频移。
5. 描述单脉冲雷达技术。

Application Examples

This concluding chapter presents overviews of satellite communication systems and radar sensors with emphasis on their electromagnetic-related aspects.

10-1 Satellite Communication Systems

Today's world is connected by a vast communication network that provides a wide array of voice, data, and video services to both fixed and mobile terminals (Fig. 10-1). The viability and effectiveness of the network are attributed in large measure to the use of orbiting satellite systems that function as relay stations with wide area coverage of Earth's surface. From a geostationary orbit at 35,786 km above the equator, a satellite can view over one-third of Earth's surface and can connect any pair of points within its coverage (Fig. 10-2). The history of communication satellite engineering dates back to the late 1950s when the U.S. Navy used the moon as a passive reflector to relay low-data-rate communications between Washington D.C. and Hawaii. The first major development involving artificial Earth satellites took place in October of 1957 when the Soviet Union launched *Sputnik I* and used it for 21 days to transmit (one-way) telemetry information to a ground receiving station. This was followed by another telemetry satellite, *Explorer I*, launched by the United States in January, 1958. An important development took place in December of that year when the United States launched the *Score* satellite and used it to broadcast President Eisenhower's Christmas message, marking the first instance of two-way voice communication via an artificial satellite.

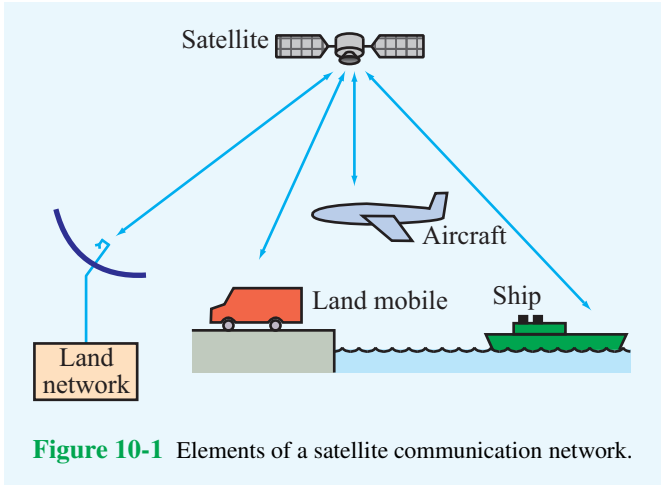


Figure 10-1 Elements of a satellite communication network.

These achievements were followed by a flurry of space activity, leading to the development of operational communication satellites by many countries for both commercial and governmental services. This section describes satellite communications links with emphasis on transmitter-receiver power calculations, propagation aspects, frequency allocations, and antenna design considerations.

A satellite is said to be in a *geostationary orbit* around the Earth when it is in a circular orbit in a plane identical with Earth's equatorial plane at an altitude where the orbital period is identical with Earth's rotational period, thereby appearing stationary relative to Earth's surface. A satellite of mass M_s in circular orbit around Earth (Fig. 10-3) is subject to two forces: the attractive gravitational force F_g and the repelling centrifugal force F_c . The magnitudes of these two forces are given by

$$F_g = \frac{GM_s M_e}{R_0^2}, \tag{10.1}$$

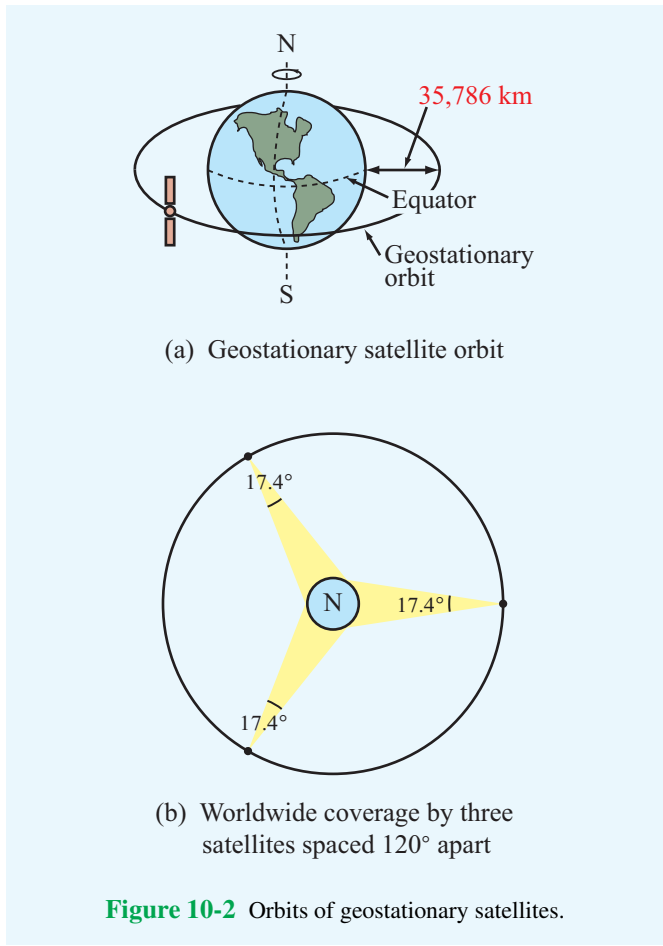


Figure 10-2 Orbits of geostationary satellites.

应用实例

最后一章概述了卫星通信系统和雷达传感器，重点介绍了它们的电磁相关方面。

10-1 卫星通信系统

当今世界通过庞大的通信网络连接起来，为固定和移动终端提供广泛的语音、数据和视频服务（图10-1）。该网络的可行性和有效性在很大程度上归功于轨道卫星系统的使用，这些系统充当中继站，广泛覆盖地球表面。从赤道上空 35,786 公里的地球静止轨道上，卫星可以看到地球表面的三分之一以上，并可以连接其覆盖范围内的任意一对点（图 10-2）。通信卫星工程的历史可以追溯到 20 世纪 50 年代末，当时美国海军利用月球作为无源反射器来中继华盛顿特区和夏威夷之间的低数据速率通信。涉及人造地球卫星的第一个重大发展发生在 1957 年 10 月，当时苏联发射了 *Sputnik I*，并使用它长达 21 天向地面接收站传输（单向）遥测信息。随后，美国于 1958 年 1 月发射了另一颗遥测卫星 *Explorer I*。同年 12 月，美国发射了 *Score* 卫星，并用它来广播艾森豪威尔总统的圣诞致辞，这是一项重要的进展，标志着通过人造卫星进行双向语音通信的首次实例。

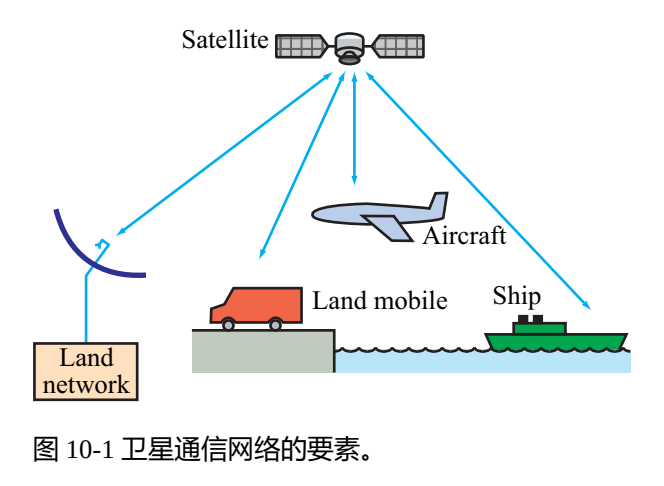


图 10-1 卫星通信网络的要素。

这些成就之后是一系列太空活动，导致许多国家开发了用于商业和政府服务的业务通信卫星。本节描述卫星通信链路，重点是发射机-接收机功率计算、传播方面、频率分配和天线设计注意事项。

当卫星处于与地球赤道平面相同的平面上的圆形轨道上且轨道周期与地球自转周期相同的高度时，卫星被称为处于绕地球地球静止轨道，因此相对于地球表面而言是静止的。质量为 M_s 的卫星在绕地球运行的圆形轨道上（图 10-3）受到两种力的作用：吸引力 F_g 和排斥离心力 F_c 。这两个力的大小由下式给出

$$F_g = \frac{GM_s M_e}{R_0^2}, \tag{10.1}$$

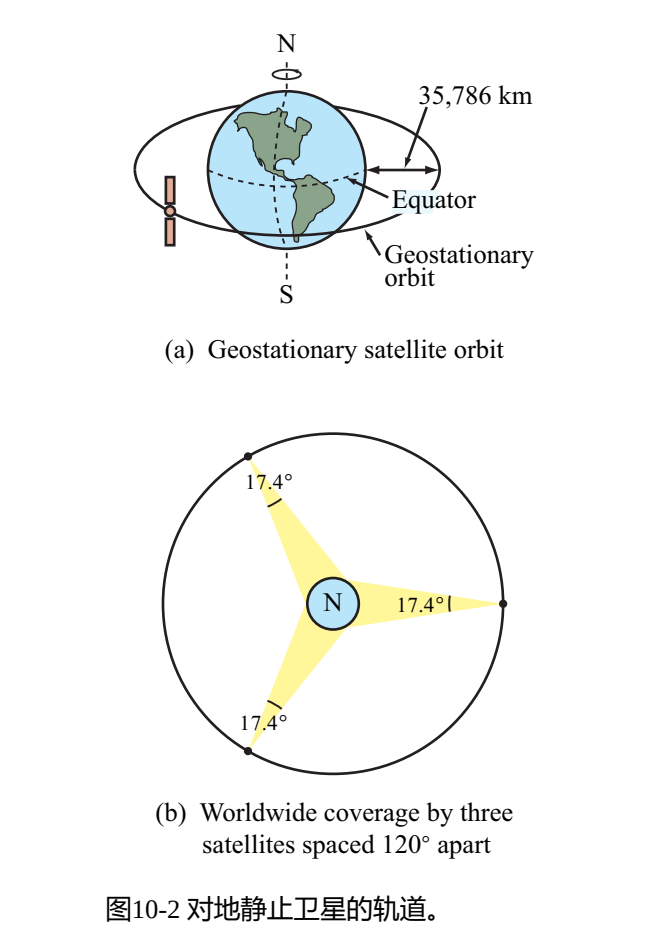


图10-2 对地静止卫星的轨道。

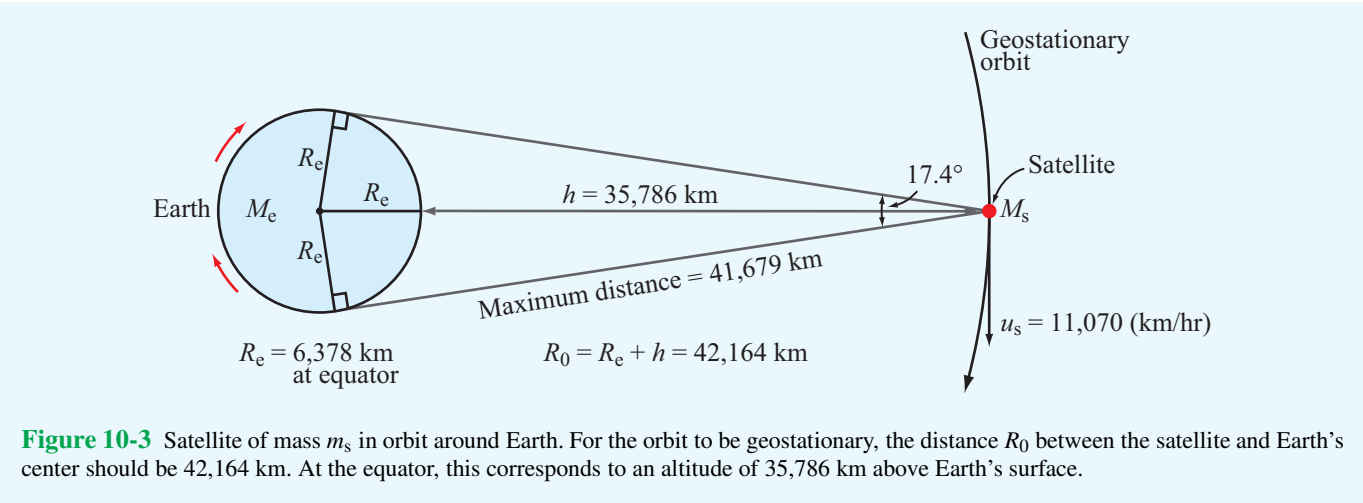


Figure 10-3 Satellite of mass m_s in orbit around Earth. For the orbit to be geostationary, the distance R_0 between the satellite and Earth's center should be 42,164 km. At the equator, this corresponds to an altitude of 35,786 km above Earth's surface.

$$F_c = \frac{M_s u_s^2}{R_0} = M_s \omega^2 R_0, \tag{10.2}$$

where $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ is the universal gravitational constant, $M_e = 5.98 \times 10^{24} \text{ kg}$ is Earth's mass, R_0 is the distance between the satellite and the center of Earth, and u_s is the satellite velocity. For a rotating object, $u_s = \omega R_0$, where ω is its angular velocity. In order for the satellite to remain in orbit, the two opposing forces acting on it have to be equal in magnitude, or

$$G \frac{M_s M_e}{R_0^2} = M_s \omega^2 R_0, \tag{10.3}$$

which yields a solution for R_0 given by

$$R_0 = \left(\frac{G M_e}{\omega^2} \right)^{1/3}. \tag{10.4}$$

To remain stationary with respect to Earth's surface, the satellite's angular velocity has to be the same as that of Earth's own angular velocity around its own axis. Thus,

$$\omega = \frac{2\pi}{T}, \tag{10.5}$$

where T is the period of one sidereal day in seconds. A sidereal day, which takes into account Earth's rotation around the sun, is equal to 23 hours, 56 minutes, and 4.1 seconds. Using Eq. (10.5) in Eq. (10.4) gives

$$R_0 = \left(\frac{G M_e T^2}{4\pi^2} \right)^{1/3}, \tag{10.6}$$

and upon using the numerical values for T , M_e , and G , we obtain the result $R_0 = 42,164 \text{ km}$. Subtracting 6,378 km for Earth's mean radius at the equator gives an altitude of $h = 35,786 \text{ km}$ above Earth's surface.

From a geostationary orbit, Earth subtends an angle of 17.4° , covering an arc of about 18,000 km along the equator, which corresponds to a longitude angle of about 160° . With three equally spaced satellites in geostationary orbit over the Earth's equator, it is possible to achieve complete global coverage of the entire equatorial plane with significant overlap between the beams of the three satellites. Concerning coverage toward the poles, a global beam can reach Earth stations up to 81° of latitude on either side of the equator.

Not all satellite communication systems use spacecraft that are in geostationary orbits. Indeed, because of transmitter power limitations or other considerations, it is sometimes necessary to operate from much lower altitudes. In this case, the satellite is placed in a **highly elliptical orbit** (to satisfy Kepler's law) so that for part of the orbit (near its perigee) it is at a range of only a few hundred kilometers from Earth's surface. Whereas only three geostationary satellites are needed to provide near-global coverage of Earth's surface, a much larger number is needed when the satellites operate from highly elliptical orbits. A good example of the latter is the Global Positioning System (GPS) described in Technology Brief 5.

10-2 Satellite Transponders

A communication satellite functions as a distant repeater; it receives **uplink** signals from Earth stations, processes the signals, and then **downlinks** (retransmits) them to their

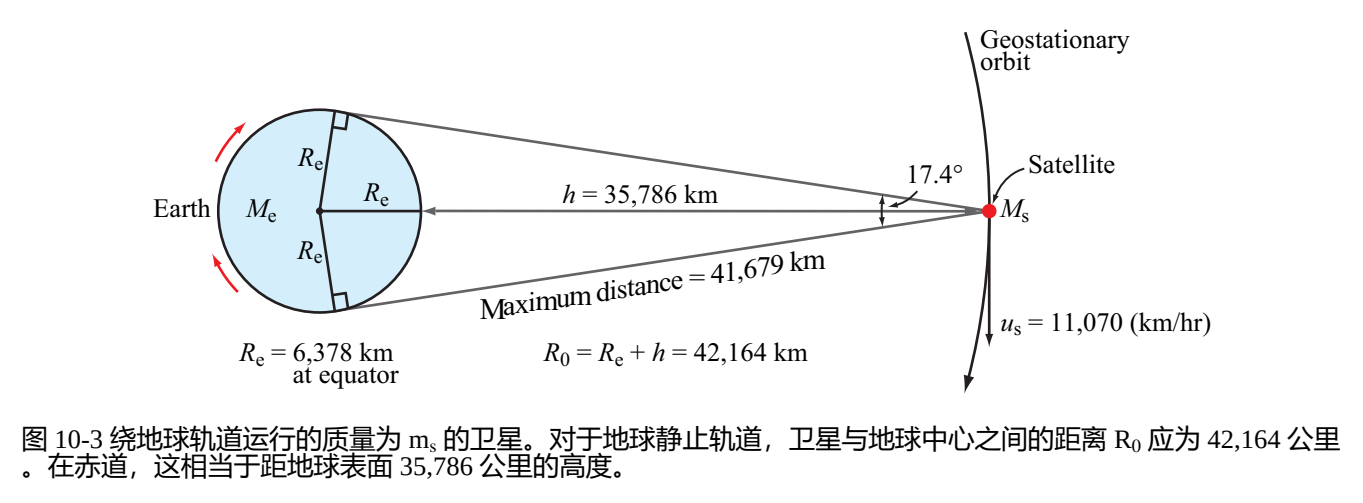


图 10-3 绕地球轨道运行的质量为 m_s 的卫星。对于地球静止轨道，卫星与地球中心之间的距离 R_0 应为 42,164 公里。在赤道，这相当于距地球表面 35,786 公里的高度。

$$F_c = \frac{M_s u_s^2}{R_0} = M_s \omega^2 R_0, \tag{10.2}$$

其中 $G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ 为万有引力常数， $M_e = 5.98 \times 10^{24} \text{ kg}$ 为地球质量， R_0 为卫星与地球中心之间的距离， u_s 为卫星速度。对于旋转物体， $u_s = \omega R_0$ ，其中 ω 是其角速度。为了使卫星保持在轨道上，作用在其上的两个相反的力必须大小相等，或者

$$G \frac{M_s M_e}{R_0^2} = M_s \omega^2 R_0, \tag{10.3}$$

得出 R_0 的解：

$$R_0 = \left(\frac{G M_e}{\omega^2} \right)^{1/3}. \tag{10.4}$$

为了相对于地球表面保持静止，卫星的角速度必须与地球自身绕其自身轴的角速度相同。因此，

$$\omega = \frac{2\pi}{T}, \tag{10.5}$$

其中 T 是一个恒星日的周期（以秒为单位）。考虑到地球绕太阳的自转，一个恒星日等于 23 小时 56 分 4.1 秒。使用方程式 (10.5) (10.4) 给出

$$R_0 = \left(\frac{G M_e T^2}{4\pi^2} \right)^{1/3}, \tag{10.6}$$

使用 T 、 M_e 和 G 的数值，我们得到结果 $R_0 = 42,164 \text{ km}$ 。减去地球赤道处的平均半径 6,378 公里，得出距地球表面 35,786 公里的高度 $h = 35,786 \text{ km}$ 。

从地球静止轨道来看，地球所成的角度为 17.4° ，沿赤道覆盖了约 18,000 公里的弧线，对应于约 160° 的经度角。利用地球赤道上空静止轨道上的三颗等距卫星，可以实现整个赤道平面的完整全球覆盖，并且三颗卫星的波束之间存在显著重叠。关于两极的覆盖范围，全球波束可以到达赤道两侧纬度高达 81° 的地球站。

并非所有卫星通信系统都使用地球静止轨道上的航天器。事实上，由于发射机功率限制或其他考虑，有时需要在低得多的高度进行操作。在这种情况下，卫星被放置在高椭圆轨道上（以满足开普勒定律），因此轨道的一部分（近地点附近）距离地球表面仅几百公里。虽然只需要三颗地球静止卫星即可提供地球表面的近乎全球覆盖，但当卫星在高椭圆轨道上运行时，需要的数量要多得多。后者的一个很好的例子是技术简介 5 中描述的全局定位系统 (GPS)。

10-2 卫星转发器

通信卫星充当远距离中继器；它从地球站接收上行信号，处理信号，然后将它们下行（重新传输）到它们的

Table 10-1 Communications satellite frequency allocations.

Use	Downlink Frequency (MHz)	Uplink Frequency (MHz)
	Fixed Service	
Commercial (C-band)	3,700–4,200	5,925–6,425
Military (X-band)	7,250–7,750	7,900–8,400
Commercial (K-band)		
Domestic (USA)	11,700–12,200	14,000–14,500
International	10,950–11,200	27,500–31,000
	Mobile Service	
Maritime	1,535–1,542.5	1,635–1,644
Aeronautical	1,543.5–1,558.8	1,645–1,660
	Broadcast Service	
	2,500–2,535	2,655–2,690
	11,700–12,750	
	Telemetry, Tracking, and Command	
	137–138, 401–402, 1,525–1,540	

intended Earth destinations. The International Telecommunication Union has allocated specific bands for satellite communications (Table 10-1). Of these, the bands used by the majority of U.S. commercial satellites for domestic communications are the *4/6 GHz band* (3.7 to 4.2 GHz downlink and 5.925 to 6.425 GHz uplink) and the *12/14 GHz band* (11.7 to 12.2 GHz downlink and 14.0 to 14.5 GHz uplink). Each uplink and downlink segment has been allocated 500 MHz of bandwidth. By using different frequency bands for Earth-to-satellite uplink segment and for satellite-to-Earth downlink segments, the same antennas can be used for both functions while simultaneously guarding against interference between the two signals. The downlink segment commonly uses a lower-frequency carrier than the uplink segment because lower frequencies suffer lower attenuation by Earth’s atmosphere, thereby easing the requirement on satellite output power.

We shall use the 4/6 GHz band as a model to discuss the satellite-repeater operation, while keeping in mind that the functional configuration of the repeater is basically the same regardless of which specific communication band is used.

Figure 10-4 shows a generalized block diagram of a typical 12-channel repeater. The path of each channel—from the point of reception by the antenna, transfer through the repeater, and final retransmission through the antenna—is called a *transponder*. The available 500 MHz bandwidth is allocated to 12 channels (transponders) of 36 MHz bandwidth per channel and 4 MHz separation between channels. The basic

functions of a transponder are (a) isolation of neighboring radio frequency (RF) channels, (b) frequency translation, and (c) amplification. With *frequency-division multiple access (FDMA)*—one of the schemes commonly used for information transmission—each transponder can accommodate thousands of individual telephone channels within its 36 MHz of bandwidth (telephone speech signals require a minimum bandwidth of 3 kHz, so frequency spacing is nominally 4 kHz per telephone channel), several TV channels (each requiring a bandwidth of 6 MHz), millions of bits of digital data, or combinations of all three.

When the same antenna is used for both transmission and reception, a *duplexer* is used to perform the signal separation. Many types of duplexers are available, but among the simplest to understand is the circulator shown in Fig. 10-5. A *circulator* is a three-port device that uses a ferrite material placed in a magnetic field induced by a permanent magnet to achieve power flow from ports 1 to 2, 2 to 3, and 3 to 1, but not in the reverse directions. With the antenna connected to port 1, the received signal is channeled only to port 2; if port 2 is properly matched to the band-pass filter, no part of the received signal is reflected from port 2 to 3. Similarly, the transmitted signal connected to port 3 is channeled by the circulator to port 1 for transmission by the antenna.

Following the duplexer shown in Fig. 10-4, the received signal passes through a receiver bandpass filter that ensures isolation of the received signal from the transmitted signal. The receiver filter covers the bandwidth from 5.925 to 6.425 GHz, which encompasses the cumulative bandwidths of all 12 channels; the first received channel extends from 5,927 to 5,963 MHz, the second one from 5,967 to 6,003 MHz, and so on until the twelfth channel, which covers the range from 6,367 to 6,403 MHz. Tracing the signal path, the next subsystem is the wideband receiver, which consists of three elements: a low-noise wideband amplifier, a frequency translator, and an output amplifier. The frequency translator consists of a stable local oscillator, which generates a signal at frequency $f_0 = 10,105$ MHz, connected to a nonlinear microwave mixer. The mixer serves to convert the frequency f_r of the received signal (which covers the range from 5,927 to 6,403 MHz) to a lower-frequency signal $f_i = f_0 - f_r$. Thus, the lower end of the received signal frequency band gets converted from 5,927 to 4,178 MHz and the upper end gets converted from 6,403 to 3,702 MHz. This translation results in 12 channels with new frequency ranges, but these signals carry the same information (modulation) that was present in the received signals. In principle, the receiver output signal can now be further amplified and then channeled to the antenna through the duplexer for transmission back to Earth. Instead, the receiver output signal is separated into the 12 transponder channels through a *multiplexer* followed by a bank of narrow band-pass

表10-1 通信卫星频率分配。

Use	Downlink Frequency (MHz)	Uplink Frequency (MHz)
	Fixed Service	
Commercial (C-band)	3,700–4,200	5,925–6,425
Military (X-band)	7,250–7,750	7,900–8,400
Commercial (K-band)		
Domestic (USA)	11,700–12,200	14,000–14,500
International	10,950–11,200	27,500–31,000
	Mobile Service	
Maritime	1,535–1,542.5	1,635–1,644
Aeronautical	1,543.5–1,558.8	1,645–1,660
	Broadcast Service	
	2,500–2,535	2,655–2,690
	11,700–12,750	
	Telemetry, Tracking, and Command	
	137–138, 401–402, 1,525–1,540	

预定的地球目的地。国际电信联盟为卫星通信分配了特定频段（表10-1）。其中，大多数美国商业卫星用于国内通信的频段是4/6 GHz频段（3.7至4.2 GHz下行链路和5.925至6.425 GHz上行链路）和12/14 GHz频段（11.7至12.2 GHz下行链路和14.0至14.5 GHz上行链路）。每个上行链路和下行链路段被分配了500 MHz的带宽。通过对地球到卫星上行链路段和卫星到地球下行链路段使用不同的频段，相同的天线可以用于这两种功能，同时防止两个信号之间的干扰。下行链路段通常使用比上行链路段更低频率的载波，因为较低频率受到地球大气层的衰减较小，从而减轻了对卫星输出功率的要求。

我们将使用 4/6 GHz 频段作为模型来讨论卫星中继器的操作，同时记住，无论使用哪个特定通信频段，中继器的功能配置基本相同。

图 10-4 显示了典型 12 通道中继器的通用框图。每个通道的路径（从天线接收点，通过中继器传输，最后通过天线重传）称为转发器。可用的 500 MHz 带宽分配给 12 个通道（转发器），每个通道 36 MHz 带宽，通道之间间隔 4 MHz。基本的

应答器的功能包括 (a) 相邻射频 (RF) 通道的隔离、(b) 频率转换和 (c) 放大。通过频分多址 (FDMA)（信息传输常用的方案之一），每个转发器可以在其 36 MHz 带宽内容纳数千个单独的电话通道（电话语音信号需要的最小带宽为 3 kHz，因此每个电话通道的频率间隔名义上为 4 kHz）、多个电视通道（每个通道需要 6 MHz 的带宽）、数百万比特的数字数据或三者的组合。

当同一天线用于发送和接收时，使用双工器来执行信号分离。双工器有多种类型，但最容易理解的是图 10-5 所示的循环器。环形器是一种三端口器件，它使用放置在永磁体感应磁场中的铁氧体材料来实现从端口 1 到 2、2 到 3 和 3 到 1 的功率流，但不能反向流动。当天线连接到端口 1 时，接收到的信号仅传输到端口 2；如果端口 2 与带通滤波器正确匹配，接收信号的任何部分都不会从端口 2 反射到端口 3。同样，连接到端口 3 的发射信号由循环器引导到端口 1，以便通过天线进行发射。

在图 10-4 所示的双工器之后，接收信号通过接收器带通滤波器，确保接收信号与发射信号的隔离。接收器滤波器覆盖 5.925 至 6.425 GHz 的带宽，其中包含所有 12 个通道的累积带宽；第一个接收信道从 5,927 MHz 延伸到 5,963 MHz，第二个接收信道从 5,967 MHz 延伸到 6,003 MHz，以此类推，直到第十二个信道，覆盖从 6,367 MHz 到 6,403 MHz 的范围。沿着信号路径，下一个子系统是宽带接收器，它由三个元件组成：低噪声宽带放大器、频率转换器和输出放大器。频率转换器由一个稳定的本地振荡器组成，该振荡器生成频率为 $f_0 = 10,105$ MHz 的信号，并连接到非线性微波混频器。混频器用于将接收信号的频率 f_r （覆盖范围为 5,927 至 6,403 MHz）转换为低频信号 $f_i = f_0 - f_r$ 。因此，接收信号频带的下端从 5,927 MHz 转换为 4,178 MHz，上端从 6,403 MHz 转换为 3,702 MHz。这种转换会产生具有新频率范围的 12 个通道，但这些信号携带与接收信号中存在的相同信息（调制）。原则上，接收器输出信号现在可以进一步放大，然后通过双工器传输到天线，然后传输回地球。相反，接收器输出信号通过复用器分为 12 个应答器通道，然后是一组窄带通

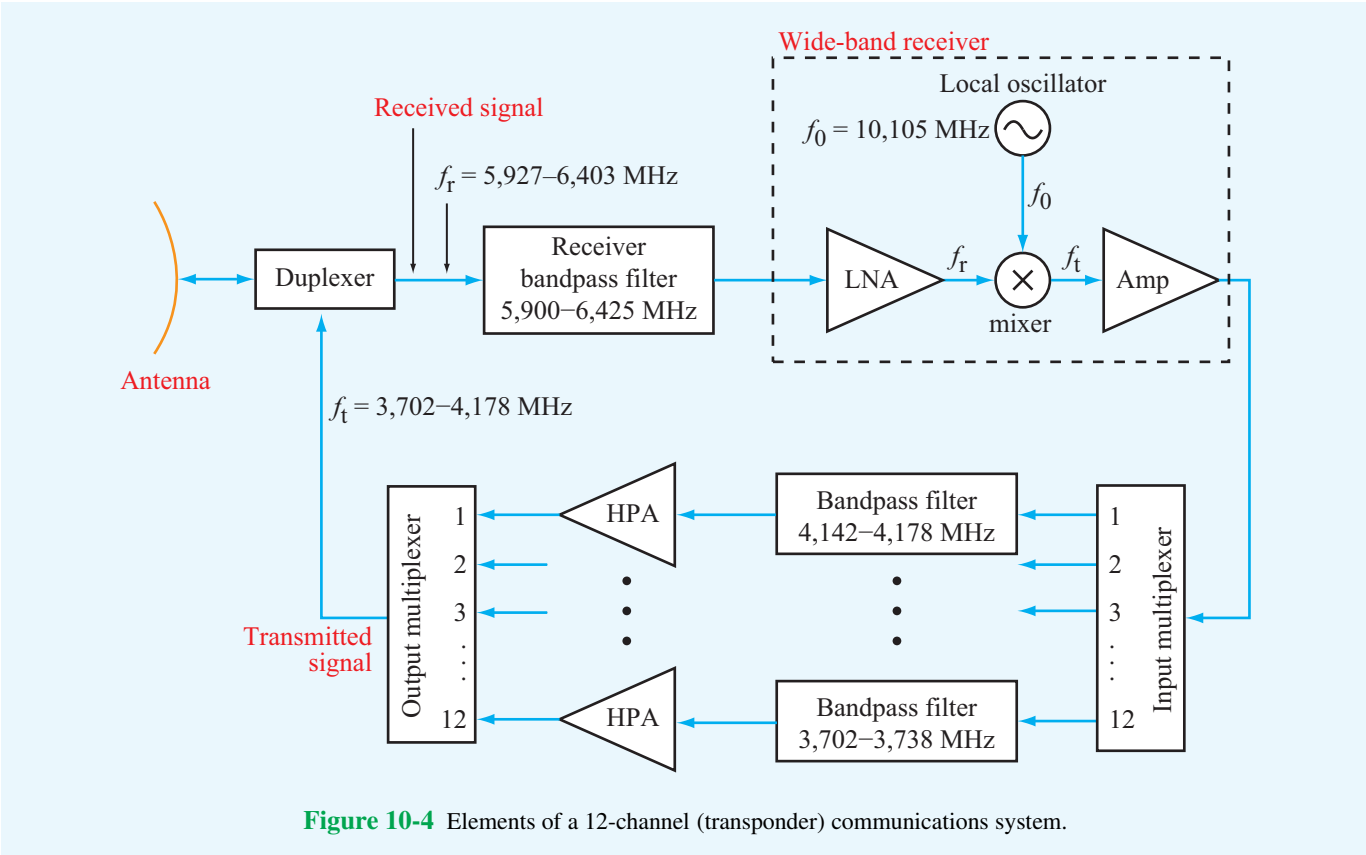


Figure 10-4 Elements of a 12-channel (transponder) communications system.

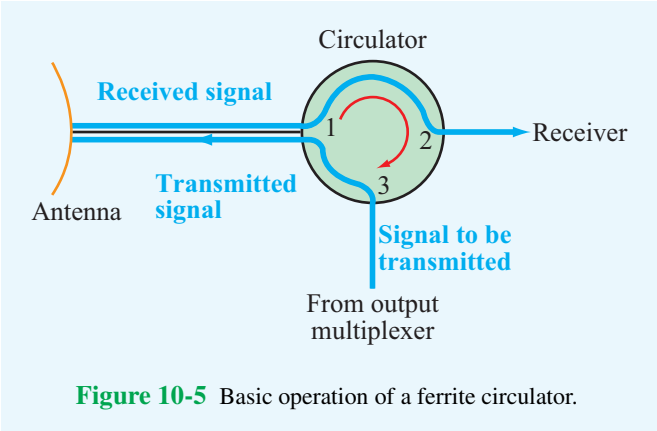


Figure 10-5 Basic operation of a ferrite circulator.

by another multiplexer that feeds the combined spectrum into the duplexer. This channel separation and recombination process is used as a safety measure against losing all 12 channels should a high-power amplifier experience total failure or degradation in performance.

The information carrying capacity of a satellite repeater can be doubled from 12 to 24 channels over the same 500 MHz bandwidth by using **polarization diversity**. Instead of transmitting one channel of information over channel 1 (5,927 to 5,963 MHz), for example, the ground station transmits to the satellite two signals carrying different information and covering the same frequency band, but with different antenna polarization configurations, such as right-hand circular (RHC) and left-hand circular (LHC) polarizations. The satellite antenna is equipped with a feed arrangement that can receive each of the two circular polarization signals individually with negligible interference between them. Two duplexers are used in this case: one connected to the RHC polarization feed and another connected to the LHC polarization feed, as illustrated in Fig. 10-6.

filters with each covering the bandwidth of one transponder channel. Each of the 12 channels is amplified by its own **high-power amplifier (HPA)**, and then the 12 channels are combined

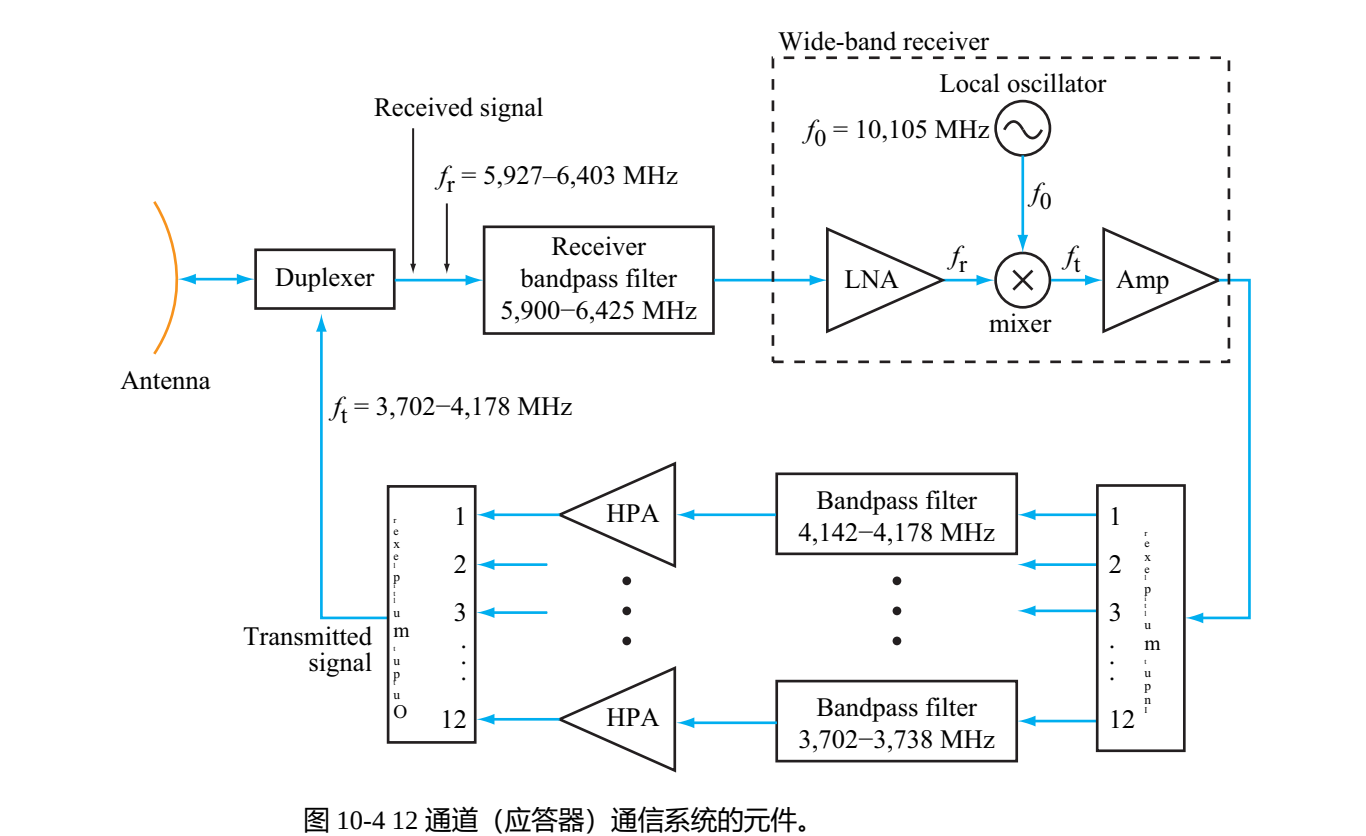


图 10-4 12 通道 (应答器) 通信系统的元件。

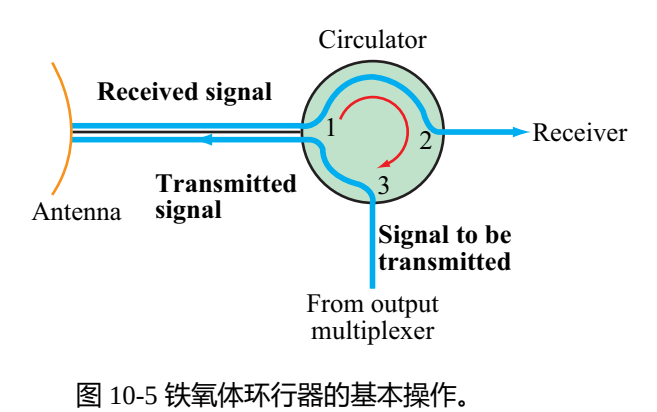


图 10-5 铁氧体环行器的基本操作。

每个滤波器都覆盖一个转发器通道的带宽。12 个通道中的每一个通道均由其自己的高功率放大器 (HPA) 进行放大, 然后将 12 个通道组合起来

由另一个多路复用器将组合频谱馈入双工器。这种通道分离和重组过程被用作一种安全措施, 防止在高功率放大器出现完全故障或性能下降时丢失所有 12 个通道。

通过使用极化分集, 卫星中继器的信息承载能力可以在相同的 500 MHz 带宽上从 12 个通道增加一倍到 24 个通道。例如, 地面站不是通过信道 1 (5,927 至 5,963 MHz) 传输一个信道的信息, 而是向卫星发送两个携带不同信息并覆盖相同频段的信号, 但具有不同的天线极化配置, 例如右旋圆 (RHC) 和左旋圆 (LHC) 极化。卫星天线配备了馈电装置, 可以单独接收两个圆极化信号中的每一个, 并且它们之间的干扰可以忽略不计。在这种情况下使用两个双工器: 一个连接到 RHC 偏振馈电, 另一个连接到 LHC 偏振馈电, 如图 10-6 所示。

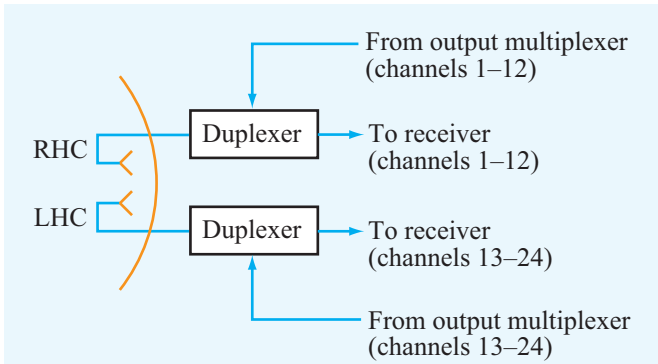


Figure 10-6 Polarization diversity is used to increase the number of channels from 12 to 24.

10-3 Communication-Link Power Budget

The uplink and downlink segments of a satellite communication link (Fig. 10-7) are each governed by the *Friis transmission formula* (Section 9-6), which states that the power P_r received by an antenna with gain G_r due to the transmission of power P_t by an antenna with gain G_t at a range R is given by

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.7)$$

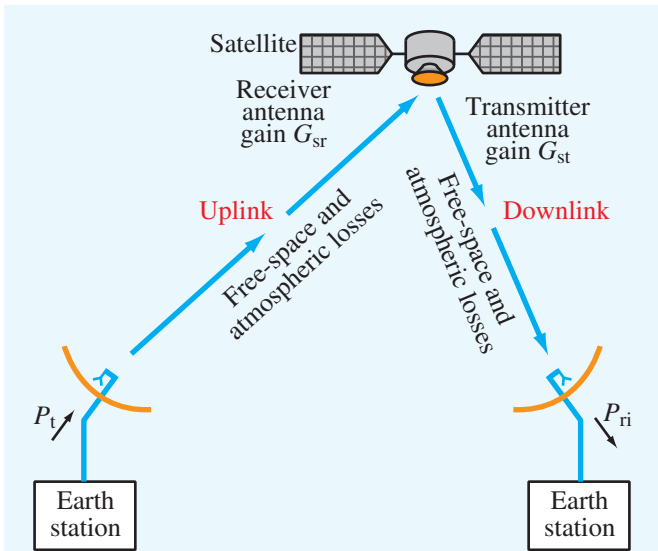


Figure 10-7 Satellite transponder.

This expression applies to a lossless medium, such as free space. To account for attenuation by clouds and rain in Earth's atmosphere (when present along the propagation path), as well as absorption by certain atmospheric gases (primarily oxygen and water vapor), we rewrite Eq. (10.7) as

$$P_{ri} = \Upsilon(\theta) P_r = \Upsilon(\theta) P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.8)$$

Now, P_{ri} represents the input power at the receiver with atmospheric losses taken into account, and $\Upsilon(\theta)$ is the one-way *transmissivity* of the atmosphere at zenith angle θ . In addition to its dependence on θ , $\Upsilon(\theta)$ is a function of the frequency of the communication link and the rain-rate conditions along the propagation path. At frequencies below 10 GHz, which include the 4/6 GHz band allocated for satellite communication, absorption by atmospheric gases is very small, as is attenuation due to clouds and rain. Consequently, the magnitude of $\Upsilon(\theta)$ is typically on the order of 0.5 to 1 for most conditions. A transmissivity of 0.5 means that twice as much power needs to be transmitted (compared to the free-space case) in order to receive a specified power level. Among the various sources of atmospheric attenuation, the most serious is rainfall, and its attenuation coefficient increases rapidly with increasing frequency. Consequently, atmospheric attenuation assumes greater importance with regard to transmitter power requirements as the communication-system frequency is increased toward higher bands in the microwave region.

The noise appearing at the receiver output, P_{no} , consists of three contributions: (1) noise internally generated by the receiver electronics, (2) noise picked up by the antenna due to external sources, including emission by the atmosphere, and (3) noise due to thermal emission by the antenna material. The combination of all noise sources can be represented by an equivalent *system noise temperature*, T_{sys} , defined such that

$$P_{no} = G_{rec} K T_{sys} B, \quad (10.9)$$

where K is Boltzmann's constant, and G_{rec} and B are the receiver power gain and bandwidth, respectively. This output noise level is the same as would appear at the output of a noise-free receiver with input noise level:

$$P_{ni} = \frac{P_{no}}{G_{rec}} = K T_{sys} B. \quad (10.10)$$

The *signal-to-noise ratio* is defined as the ratio of the signal power to the noise power *at the input of an equivalent noise-free receiver*. Hence,

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta) P_t G_t G_r}{K T_{sys} B} \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.11)$$

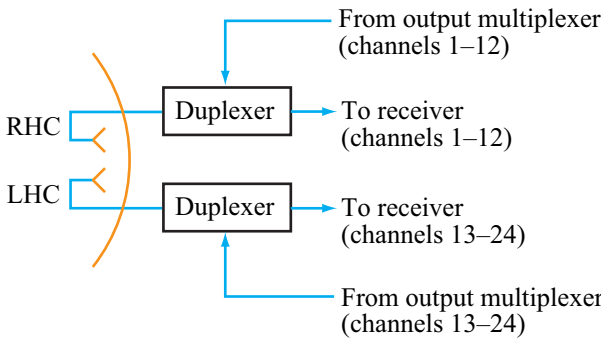


图10-6 极化分集用于将通道数从12个增加到24个。

10-3 通信链路功率预算

卫星通信链路的上行链路和下行链路段 (图 10-7) 均由 Friis 传输公式 (第 9-6 节) 控制, 该公式指出, 由于增益为 G_t 的天线在范围 R 上发射功率 P_t , 增益为 G_r 的天线接收到的功率 P_r 由下式给出:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.7)$$

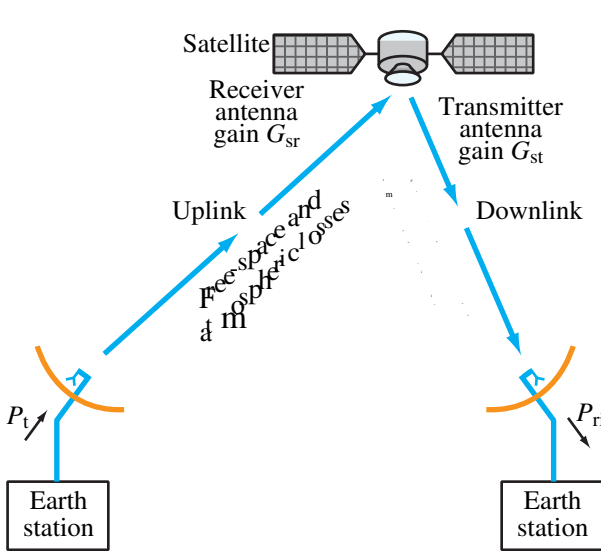


图 10-7 卫星转发器。

此表达式适用于无损介质, 例如自由空间。为了考虑地球大气中云和雨的衰减 (当沿着传播路径存在时), 以及某些大气气体 (主要是氧气和水蒸气) 的吸收, 我们重写了方程: (10.7) 作为

$$P_{ri} = \Upsilon(\theta) P_r = \Upsilon(\theta) P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.8)$$

现在, P_{ri} 表示考虑了大气损耗的接收器处的输入功率, 而 $\Upsilon(\theta)$ 是天顶角 θ 处大气的单向透射率。除了依赖于 θ 之外, $\Upsilon(\theta)$ 也是通信链路频率和传播路径上降雨率条件的函数。在 10 GHz 以下的频率 (包括分配给卫星通信的 4/6 GHz 频段), 大气气体的吸收非常小, 云和雨造成的衰减也非常小。因此, 在大多数情况下, $\Upsilon(\theta)$ 的大小通常约为 0.5 到 1。0.5 的传输率意味着需要传输两倍的功率 (与自由空间情况相比) 才能接收指定的功率电平。在大气衰减的各种来源中, 最严重的是降雨, 其衰减系数随着频率的增加而迅速增加。因此, 随着通信系统频率向微波区域的更高频段增加, 大气衰减对于发射机功率要求变得更加重要。

接收器输出处出现的噪声 P_{no} 由三个部分组成: (1) 接收器电子设备内部产生的噪声; (2) 由于外部源 (包括大气发射) 而由天线拾取的噪声; (3) 由于天线材料的热发射而产生的噪声。所有噪声源的组合可以用等效系统噪声温度 T_{sys} 表示, 定义为

$$P_{no} = G_{rec} K T_{sys} B, \quad (10.9)$$

其中 K 是玻尔兹曼常数, G_{rec} 和 B 分别是接收器功率增益和带宽。该输出噪声电平与具有输入噪声电平的无噪声接收器的输出端出现的噪声电平相同:

$$P_{ni} = \frac{P_{no}}{G_{rec}} = K T_{sys} B. \quad (10.10)$$

信噪比定义为等效无噪声接收器输入端的信号功率与噪声功率之比。因此,

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta) P_t G_t G_r}{K T_{sys} B} \left(\frac{\lambda}{4\pi R} \right)^2. \quad (10.11)$$

The performance of a communication system is governed by two sets of issues. The first encompasses the signal-processing techniques used to encode, modulate, combine, and transmit the signal at the transmitter end and to receive, separate, demodulate, and decode the signal at the receiver end. The second set encompasses the gains and losses in the communication link, and they are represented by the signal-to-noise ratio S_n . For a given set of signal-processing techniques, S_n determines the quality of the received signal, such as the bit error rate in digital data transmission and sound and picture quality in audio and video transmissions. Very high-quality signal transmission requires very high values of S_n ; in broadcast-quality television by satellite, some systems are designed to provide values of S_n exceeding 50 dB (or a factor of 10^5).

The performance of a satellite link depends on the composite performance of the uplink and downlink segments. If either segment performs poorly, the composite performance will be poor, regardless of how good the performance of the other segment is.

10-4 Antenna Beams

Whereas most Earth-station antennas are designed to provide highly directive beams (to avoid interference effects), the satellite antenna system is designed to produce beams tailored to match the areas served by the satellite. For global coverage, beamwidths of 17.4° are required. In contrast, for transmission to and reception from a small area, beamwidths on the order of 1° or less may be needed (Fig. 10-8).

An antenna with a beamwidth β of 1° would produce a spot beam on Earth covering an area approximately 630 km in diameter.

Beam size has a direct connection to antenna gain and, in turn, to transmitter power requirements. Antenna gain G is related to the directivity D by $G = \xi D$, where ξ is the radiation efficiency and D is related to the beamwidth β by the approximate expression given by Eq. (9.26). For a circular beam,

$$G = \xi \frac{4\pi}{\beta^2}, \tag{10.12}$$

where β is in radians. For a lossless antenna ($\xi = 1$), a global beam with $\beta = 17.4^\circ (= 0.3 \text{ rad})$ corresponds to a gain $G = 136$ or 21.3 dB. A narrow 1° beam, on the other hand, corresponds to an antenna gain of 41,253 or 46.2 dB.

To accommodate the various communication functions associated with satellite systems, four main types of antennas are used.*

*R. G. Meadows and A. J. Parsons, *Satellite Communications*, Hutchinson Publishers, London, 1989.

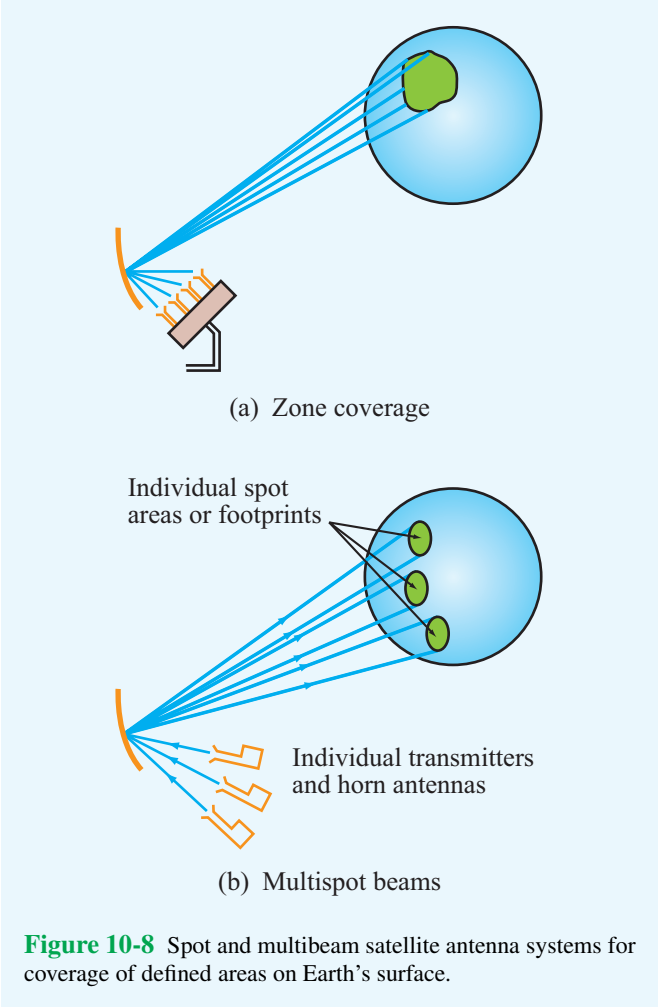


Figure 10-8 Spot and multibeam satellite antenna systems for coverage of defined areas on Earth's surface.

1. Dipoles and helices at VHF and UHF are used for telemetry, tracking, and command functions.
2. Horns and relatively small parabolic dishes (with diameters on the order of a few centimeters) to produce wide-angle beams for global coverage.
3. Parabolic dishes fed by one or more horns provide a beam for zone coverage (Fig. 10-8(a)) or multiple spot beams (Fig. 10-8(b)).
4. Antenna arrays consisting of many individual radiating elements are used to produce multispot beams and for beam steering and scanning.

通信系统的性能由两组问题决定。第一类包括信号处理技术，用于在发送端对信号进行编码、调制、组合和发送，以及在接收端接收、分离、解调和解码信号。第二组包含通信链路中的增益和损耗，它们由信噪比 S_n 表示。对于一组给定的信号处理技术， S_n 决定接收信号的质量，例如数字数据传输中的误码率以及音频和视频传输中的声音和图像质量。非常高质量的信号传输需要非常高的 S_n 值；在卫星广播质量电视中，一些系统设计为提供超过 50 dB（或 10^5 因子）的 S_n 值。

卫星链路的性能取决于上行链路和下行链路段的综合性能。如果任一部分表现不佳，则无论其他部分的表现有多好，综合表现也将很差。

10-4 天线波束

大多数地球站天线的设计目的是提供高度定向的波束（以避免干扰效应），而卫星天线系统的设计目的是产生适合卫星服务区域的波束。对于全球覆盖，需要 17.4° 的波束宽度。相反，对于小区域的传输和接收，可能需要 1° 或更小的波束宽度（图 10-8）。

波束宽度 β 为 1° 的天线将在地球上产生覆盖直径约 630 公里区域的点波束。

波束尺寸与天线增益直接相关，进而与发射机功率要求直接相关。天线增益 G 通过 $G = \xi D$ 与方向性 D 相关，其中 ξ 是辐射效率， D 通过等式给出的近似表达式与波束宽度 β 相关。（9.26）。对于圆形波束，

$$G = \xi \frac{4\pi}{\beta^2}, \tag{10.12}$$

其中 β 的单位是弧度。对于无损天线 ($\xi = 1$)， $\beta = 17.4^\circ (= 0.3 \text{ rad})$ 的全局波束对应于增益 $G = 136$ 或 21.3 dB。另一方面，窄 1° 波束对应于 41,253 或 46.2 dB 的天线增益。

为了适应与卫星系统相关的各种通信功能，使用了四种主要类型的天线。*

*R. G. Meadows and A. J. Parsons, *Satellite Communications*, Hutchinson Publishers, London, 1989.

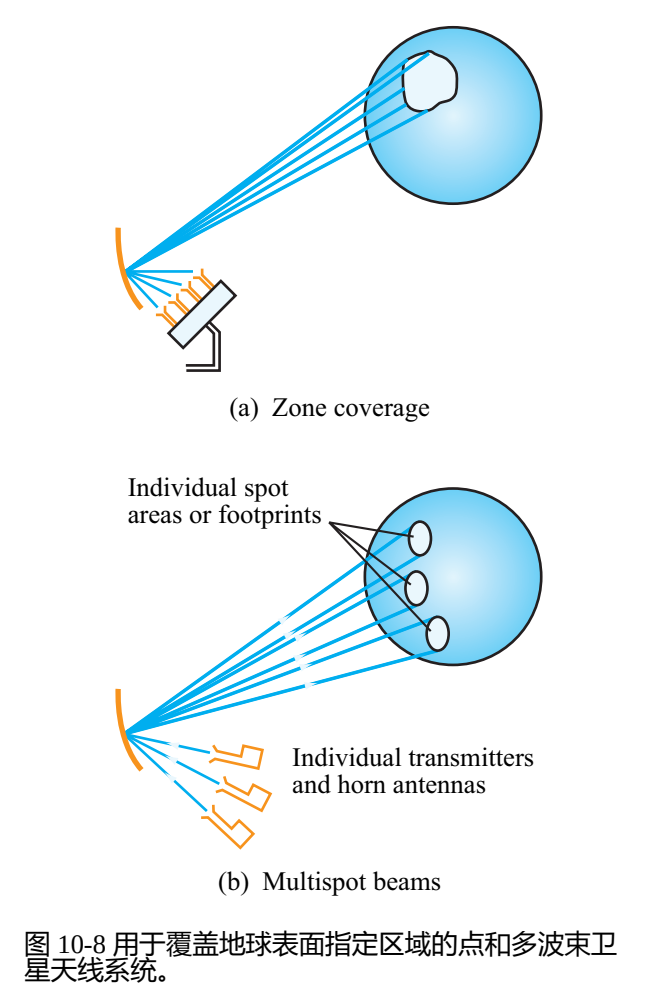


图 10-8 用于覆盖地球表面指定区域的点和多波束卫星天线系统。

1. VHF 和 UHF 的偶极子和螺旋用于遥测、跟踪和命令功能。
2. 喇叭和相对较小的抛物面天线（直径约为几厘米）可产生覆盖全球的广角波束。
3. 由一个或多个喇叭馈送的抛物面天线提供区域覆盖波束（图 10-8(a)）或多点波束（图 10-8(b)）。
4. 由许多单独的辐射元件组成的天线阵列用于产生多点波束并用于波束控制和扫描。

- Concept Question 10-1:** What are the advantages and disadvantages of elliptical satellite orbits in comparison to the geostationary orbit?
- Concept Question 10-2:** Why do satellite communication systems use different frequencies for the uplink and downlink segments? Which segment uses the higher frequency and why?
- Concept Question 10-3:** How does the use of antenna polarization increase the number of channels carried by the communication system?
- Concept Question 10-4:** What are the sources of noise that contribute to the total system noise temperature of a receiver?

and determining its *range* by measuring the time delay of short-duration pulses transmitted by the radar, a radar is also capable of specifying the *position* of the target and its radial velocity. Measurement of the *radial velocity* of a moving object is realized by measuring the *Doppler frequency shift* produced by the object. Also, the strength and shape of the reflected pulse carry information about the shape and material properties of the reflecting object.

Radar is used for a wide range of civilian and military applications, including air traffic control, aircraft navigation, law enforcement, control and guidance of weapon systems, remote sensing of Earth's environment, weather observation, astronomy, and collision avoidance for automobiles. The frequency bands used for the various types of radar applications extend from the megahertz region to frequencies as high as 225 GHz.

10-5.1 Basic Operation of a Radar System

The block diagram shown in Fig. 10-9 contains the basic functional elements of a pulse radar system. The *synchronizer-modulator* unit serves to synchronize the operation of the transmitter and the *videoprocessor-display unit*; it does so by generating a train of direct-current (dc) narrow-duration, evenly spaced pulses. These pulses, which are supplied to both the transmitter, and the videoprocessor-display unit, specify the times at which radar pulses are transmitted. The transmitter contains a high-power radio-frequency (RF) oscillator with an on/off control voltage actuated by the pulses supplied by the synchronizer-modulator unit. Hence, the transmitter generates pulses of RF energy equal in duration and spacing to the dc pulses generated by the synchronizer-modulator unit. Each pulse is supplied to the antenna through a *duplexer*, which

10-5 Radar Sensors

The term *radar* is a contracted form of the phrase *radio detection and ranging*, which conveys some—not all—of the features of a modern radar system. Historically, radar systems were first developed and used at *radio* frequencies, including the microwave band, but we now also have light radars or *lidars* that operate at optical wavelengths. Over the years, the name radar has lost its original meaning and has come to signify any *active* electromagnetic sensor that uses its own source to illuminate a region of space and then measure the echoes generated by reflecting objects contained in that region. In addition to detecting the presence of a reflecting object

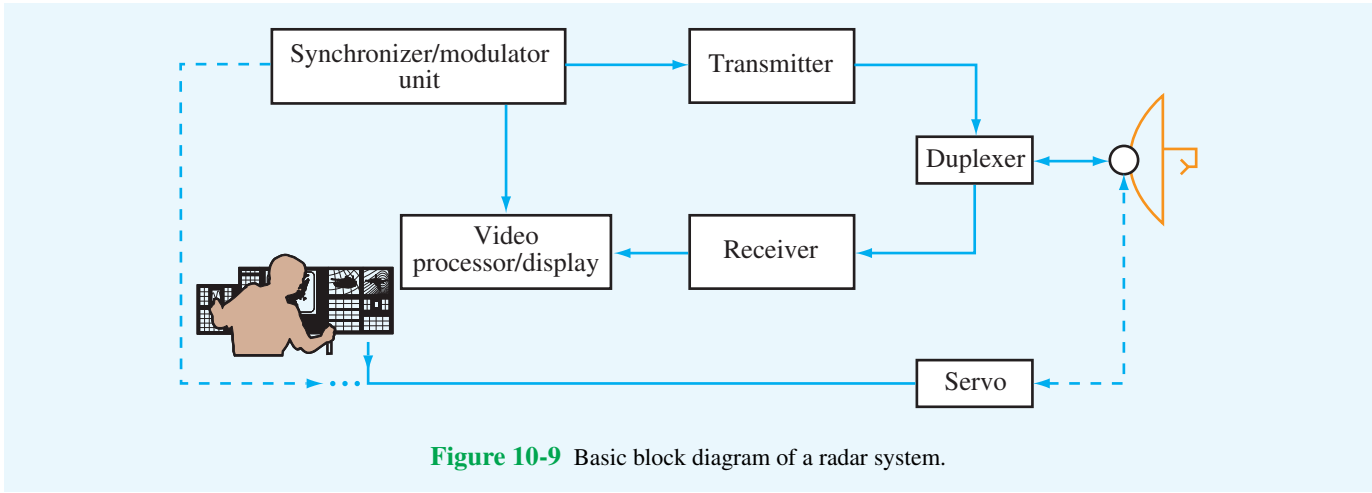


Figure 10-9 Basic block diagram of a radar system.

- Concept Question 10-1:** What are the advantages and disadvantages of elliptical satellite orbits in comparison to the geostationary orbit?
- Concept Question 10-2:** Why do satellite communication systems use different frequencies for the uplink and downlink segments? Which segment uses the higher frequency and why?
- Concept Question 10-3:** How does the use of antenna polarization increase the number of channels carried by the communication system?
- Concept Question 10-4:** What are the sources of noise that contribute to the total system noise temperature of a receiver?

通过测量雷达发射的短时脉冲的时间延迟来确定其范围，雷达还能够指定目标的位置及其径向速度。运动物体径向速度的测量是通过测量物体产生的多普勒频移来实现的。此外，反射脉冲的强度和形状还携带有关反射物体的形状和材料特性的信息。

雷达具有广泛的民用和军事应用，包括空中交通管制、飞机导航、执法、武器系统的控制和制导、地球环境遥感、天气观测、天文学和汽车防撞。用于各种类型雷达应用的频段从兆赫兹区域延伸到高达 225 GHz 的频率。

10-5.1 雷达系统的基本操作

图 10-9 所示的框图包含脉冲雷达系统的基本功能元件。同步器-调制器单元用于同步发射机和视频处理器-显示单元的操作；它通过生成一系列窄持续时间、均匀间隔的直流 (dc) 脉冲来实现这一点。这些脉冲提供给发射器和视频处理器-显示单元，指定雷达脉冲的发射时间。发射器包含一个高功率射频 (RF) 振荡器，具有由同步器-调制器单元提供的脉冲驱动的开/关控制电压。因此，发射器生成的射频能量脉冲的持续时间和间隔与同步器-调制器单元生成的直流脉冲相同。每个脉冲都通过双工器提供给天线，

10-5 雷达传感器

雷达一词是无线电探测和测距一词的缩写形式，它传达了现代雷达系统的部分（而非全部）特征。从历史上看，雷达系统首先是在射频（包括微波频段）下开发和使用的，但我们现在也有在光波长下运行的光雷达或激光雷达。多年来，雷达这个名字已经失去了它原来的含义，并开始表示任何有源电磁传感器，这些传感器使用自己的源来照亮空间区域，然后测量该区域内反射物体所产生的回波。除了检测反射物体的存在之外

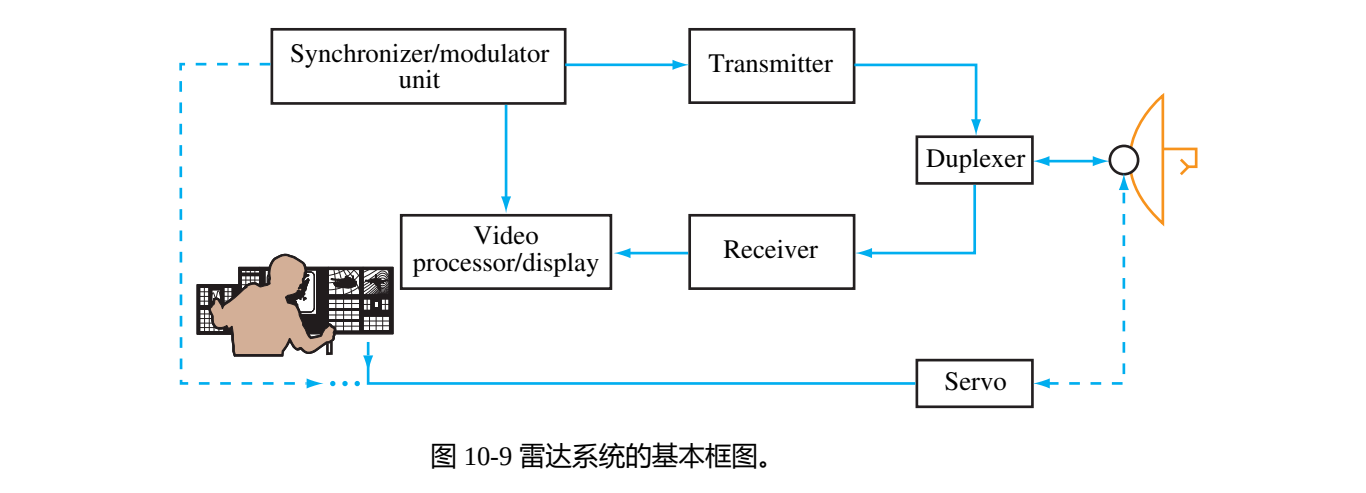


图 10-9 雷达系统的基本框图。

allows the antenna to be shared between the transmitter and the receiver. The duplexer, which often is called the **transmitter/receiver** (R) **switch**, first connects the transmitter to the antenna for the duration of the pulse and then connects the antenna to the receiver for the remaining period until the start of a new pulse. Some duplexers, however, are passive devices that perform the sharing and isolation functions continuously. The circulator shown in **Fig. 10-5** is an example of a passive duplexer. After transmission by the antenna, a portion of the transmitted signal is intercepted by a reflecting object (often called a **target**) and scattered in many directions. The energy reradiated by the target back toward the radar is collected by the antenna and delivered to the receiver, which processes the signal to detect the presence of the target and to extract information on its location and velocity. The receiver converts the reflected RF signals into lower-frequency video signals and supplies them to the videoprocessor–display unit, which displays the extracted information in a format suitable for the intended application. The servo unit positions the orientation of the antenna beam in response to control signals provided by either an operator, a control unit with preset functions, or a control unit commanded by another system. The control unit of an air-traffic-control radar, for example, commands the servo to rotate the antenna in azimuth continuously. In contrast, the radar antenna placed in the nose of an aircraft is made to scan back and forth over only a specified angular sector.

10-5.2 Unambiguous Range

The collective features of the energy transmitted by a radar are called the **signal waveform**. For a pulse radar, these features include (1) the carrier frequency f , (2) the **pulse length** τ , (3) the **pulse repetition frequency** f_p (number of pulses per second) or equivalently the **interpulse period** $T_p = 1/f_p$, and (4) the modulation (if any) within the pulses. Three of these features are illustrated in **Fig. 10-10**. Modulation, which refers

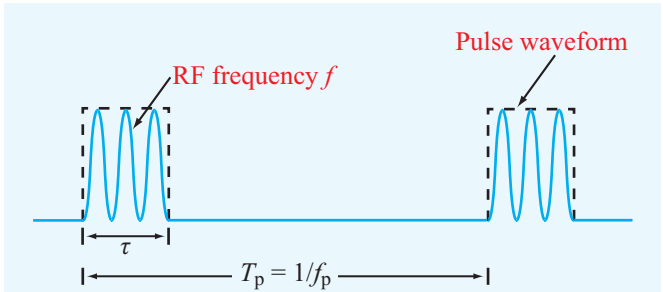


Figure 10-10 A pulse radar transmits a continuous train of RF pulses at a pulse repetition frequency f_p .

to control of the amplitude, frequency, or phase of the signal, is beyond the level of the present treatment.

The range of a target is determined by measuring the time delay T taken by the pulse to travel to the target and back. For a target at range R ,

$$T = \frac{2R}{c}, \quad (10.13)$$

where $c = 3 \times 10^8$ m/s is the speed of light and the factor 2 accounts for the two-way propagation. The maximum target range that a radar can measure unambiguously, called the **unambiguous range** R_u , is determined by the interpulse period T_p and is given by

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}. \quad (10.14)$$

The range R_u corresponds to the maximum range that a target can have such that its echo is received before the transmission of the next pulse. If T_p is too short, an echo signal due to a given pulse might arrive after the transmission of the next pulse, in which case the target would appear to be at a much shorter range than it actually is.

According to Eq. (10.14), if a radar is to be used to detect targets that are as far away as 100 km, for example, then f_p should be less than 1.5 kHz, and the higher the pulse repetition frequency (PRF), the shorter is the unambiguous range R_u . Consideration of R_u alone suggests selecting a low PRF, but other considerations suggest selecting a very high PRF. As we will see later in Section 10-6, the signal-to-noise ratio of the radar receiver is directly proportional to f_p . Hence, it would be advantageous to select a PRF as high as possible. Moreover, in addition to determining the maximum unambiguous range R_u , the PRF also determines the maximum Doppler frequency (hence, the target's maximum radial velocity) that the radar can measure unambiguously. If the requirements on maximum range and velocity cannot be met by the same PRF, then some compromise may be necessary. Alternatively, it is possible to use a **multiple-PRF** radar system that transmits a few pulses at one PRF followed by another series of pulses at another PRF. Then the two sets of received pulses are processed together to remove the ambiguities that would have been present with either PRF alone.

10-5.3 Range and Angular Resolutions

Consider a radar observing two targets located at ranges R_1 and R_2 , as shown in **Fig. 10-11**. Let $t = 0$ denote the time corresponding to the start of the transmitted pulse. The pulse length is τ . The return due to target 1 will arrive at $T_1 = 2R_1/c$ and will have a length τ (assuming that the pulse length in

允许发射器和接收器之间共享天线。双工器通常称为发射器/接收器 (R) 开关，首先在脉冲持续时间内将发射器连接到天线，然后在剩余时间内将天线连接到接收器，直到新脉冲开始。然而，一些双工器是连续执行共享和隔离功能的无源设备。图10-5所示的循环器是无源双工器的示例。通过天线传输后，部分传输信号被反射物体（通常称为目标）拦截并向多个方向散射。目标向雷达再辐射的能量由天线收集并传送到接收器，接收器处理信号以检测目标的存在并提取有关其位置和速度的信息。接收器将反射的射频信号转换为低频视频信号，并将其提供给视频处理器-显示单元，该单元以适合预期应用的格式显示提取的信息。伺服单元响应于操作员、具有预设功能的控制单元或由另一系统命令的控制单元提供的控制信号来定位天线波束的方向。例如，空中交通管制雷达的控制单元命令伺服系统连续旋转天线的方位角。相比之下，放置在飞机机头的雷达天线只能在指定的角度区域内来回扫描。

10-5.2 明确范围

雷达发射的能量的集体特征称为信号波形。对于脉冲雷达，这些特征包括 (1) 载波频率 f 、(2) 脉冲长度 τ 、(3) 脉冲重复频率 f_p （每秒脉冲数）或等效的脉冲间周期 $T_p = 1/f_p$ ，以及 (4) 脉冲内的调制（如果有）。图 10-10 展示了其中三个特征。调制，指的是

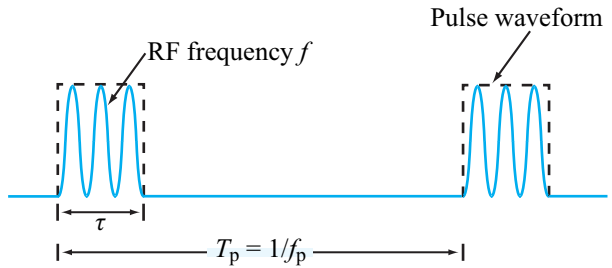


图 10-10 脉冲雷达以脉冲重复频率 f_p 发射连续的射频脉冲。

控制信号的幅度、频率或相位超出了现有治疗的水平。

目标的范围是通过测量脉冲传播到目标并返回的时间延迟 T 来确定的。对于 R 范围内的目标，

$$T = \frac{2R}{c}, \quad (10.13)$$

其中 $c = 3 \times 10^8$ m/s 是光速，因子 2 说明双向传播。雷达可以明确测量的最大目标距离，称为明确距离 R_u ，由脉冲间周期 T_p 确定，并由下式给出

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}. \quad (10.14)$$

范围 R_u 对应于目标可以具有的最大范围，以便在发送下一个脉冲之前接收到其回波。如果 T_p 太短，则给定脉冲产生的回波信号可能会在下一个脉冲传输后到达，在这种情况下，目标看起来会比实际距离短得多。

根据方程。根据式 (10.14)，如果雷达要用于探测远至100公里的目标，则 f_p 应小于1.5kHz，并且脉冲重复频率（PRF）越高，明确距离 R_u 越短。单独考虑 R_u 建议选择较低的 PRF，但其他考虑因素建议选择非常高的 PRF。正如我们稍后将在第 10-6 节中看到的，雷达接收器的信噪比与 f_p 成正比。因此，选择尽可能高的 PRF 是有利的。此外，除了确定最大明确范围 R_u 之外，PRF 还确定雷达可以明确测量的最大多普勒频率（因此，目标的最大径向速度）。如果相同的 PRF 无法满足最大范围和速度的要求，则可能需要进行一些折衷。或者，可以使用多 PRF 雷达系统，该系统在一个 PRF 处发射几个脉冲，然后在另一个 PRF 处发射另一系列脉冲。然后，对两组接收到的脉冲进行一起处理，以消除单独使用任一 PRF 时可能出现的模糊性。

10-5.3 距离和角度分辨率

考虑雷达观测位于 R_1 和 R_2 范围内的两个目标，如图 10-11 所示。令 $t = 0$ 表示对应于发射脉冲开始的时间。脉冲长度为 τ 。目标 1 的返回将到达 $T_1 = 2R_1/c$ ，并且将具有长度 τ （假设脉冲长度为

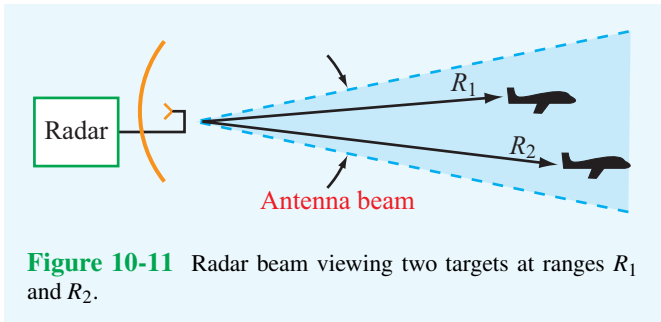


Figure 10-11 Radar beam viewing two targets at ranges R_1 and R_2 .

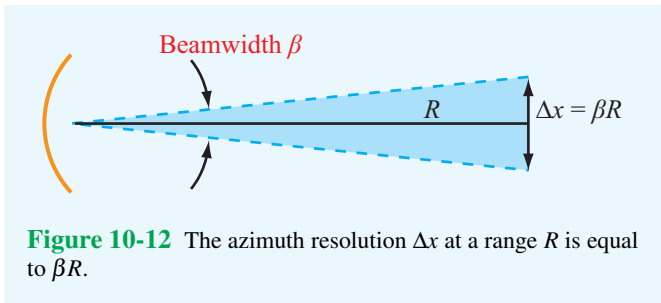


Figure 10-12 The azimuth resolution Δx at a range R is equal to βR .

space is much greater than the radial extent of the target). Similarly, the return due to target 2 will arrive at $T_2 = 2R_2/c$. The two targets are resolvable as distinct targets so long as $T_2 \geq T_1 + \tau$ or, equivalently,

$$\frac{2R_2}{c} \geq \frac{2R_1}{c} + \tau. \quad (10.15)$$

The **range resolution** of the radar, ΔR , is defined as the minimum spacing between two targets necessary to avoid overlap between the echoes from the two targets. From Eq. (10.15), this occurs when

$$\Delta R = R_2 - R_1 = c\tau/2. \quad (10.16)$$

Some radars are capable of transmitting pulses as short as 1 ns in duration or even shorter. For $\tau = 1$ ns, $\Delta R = 15$ cm.

The basic angular resolution of a radar system is determined by its antenna beamwidth β , as shown in Fig. 10-12. The corresponding **azimuth resolution** Δx at a range R is given by

$$\Delta x = \beta R, \quad (10.17)$$

where β is in radians. In some cases, special techniques are used to improve the angular resolution down to a fraction of the beamwidth. One example is the monopulse tracking radar described in Section 10-8.

10-6 Target Detection

Target detection by radar is governed by two factors: (1) the **signal energy** received by the radar receiver due to reflection of part of the transmitted energy by the target and (2) the **noise energy** generated by the receiver. Figure 10-13 depicts the output response of a radar receiver as a function of time and shows the signals due to two targets displayed against the noise contributed by external sources as well as by the devices making up the receiver. The random variations exhibited by the noise may at times make it difficult to distinguish the signal reflected by the target from a noise spike. In Fig. 10-13, the mean noise-power level at the receiver output is denoted by $P_{no} = G_{rec}P_{ni}$, where G_{rec} is the receiver gain and P_{ni} is the noise level referred to the receiver's input terminals. The power levels P_{r1} and P_{r2} represent the echoes of the two targets observed by the radar. Because of the random nature of noise, it is necessary to set a threshold level, P_{min} , for detection. For threshold level 1 indicated in Fig. 10-13, the radar will produce the presence of both targets, but it will also detect a **false alarm**. The chance of this occurring is called the **false-alarm probability**. On the other hand, if the threshold level is raised to level 2 to avoid the false alarm, the radar will not detect the presence of the first target. A radar's ability to detect the presence of a target is characterized by a **detection probability**. The setting of the threshold signal level relative to the mean noise level is thus made on the basis of a compromise that weighs both probabilities.

To keep the noise level at a minimum, the receiver is designed so that its bandwidth B is barely wide enough to pass most of the energy contained in the received pulse. Such a design, called a **matched filter**, requires that B be equal to the reciprocal of the pulse length τ (i.e., $B = 1/\tau$). Hence, for a matched-filter receiver, Eq. (10.10) becomes

$$P_{ni} = KT_{sys}B = \frac{KT_{sys}}{\tau}. \quad (10.18)$$

The signal power received by the radar, P_r , is related to the transmitted power level, P_t , through the radar equation. We will first derive the radar equation for the general case of a **bistatic radar** configuration in which the transmitter and receiver are not necessarily at the same location, and then we will specialize the results to the **monostatic radar** case where the transmitter and receiver are colocated. In Fig. 10-14, the target is at range R_t from the transmitter and at range R_r from the receiver. The power density illuminating the target is given by

$$S_t = \frac{P_t}{4\pi R_t^2} G_t \quad (\text{W/m}^2), \quad (10.19)$$

where $(P_t/4\pi R_t^2)$ represents the power density that would have been radiated by an isotropic radiator, and G_t is the gain of the

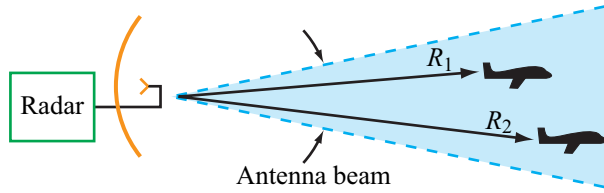


图 10-11 雷达波束观察距离 R_1 和 R_2 处的两个目标。

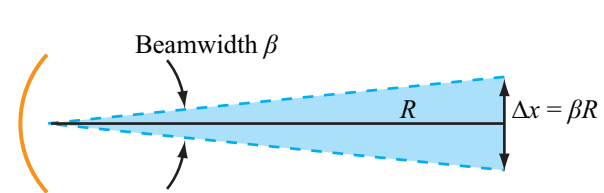


图 10-12 距离 R 处的方位角分辨率 Δx 等于 βR 。

空间远大于目标的径向范围)。类似地，目标 2 的返回将到达 $T_2 = 2R_2/c$ 。只要 $T_2 \geq T_1 + \tau$ 或等价地，这两个目标就可以解析为不同的目标

$$\frac{2R_2}{c} \geq \frac{2R_1}{c} + \tau. \quad (10.15)$$

雷达的距离分辨率 ΔR 被定义为避免两个目标的回波重叠所需的两个目标之间的最小间距。从方程。(10.15)，这发生在

$$\Delta R = R_2 - R_1 = c\tau/2. \quad (10.16)$$

有些雷达能够发射持续时间短至 1 ns 甚至更短的脉冲。对于 $\tau = 1$ ns, $\Delta R = 15$ cm。

雷达系统的基本角分辨率由其天线波束宽度 β 决定，如图 10-12 所示。范围 R 处相应的方位角分辨率 Δx 由下式给出

$$\Delta x = \beta R, \quad (10.17)$$

其中 β 的单位是弧度。在某些情况下，可以使用特殊技术将角分辨率提高到波束宽度的一小部分。一个例子是第 10-8 节中描述的单脉冲跟踪雷达。

10-6 目标检测

雷达的目标检测受两个因素控制：(1) 雷达接收器由于目标反射部分发射能量而接收到的信号能量；(2) 接收器产生的噪声能量。图 10-13 描述了雷达接收器随时间变化的输出响应，并显示了两个目标产生的信号，以及外部源以及构成接收器的设备产生的噪声。噪声表现出的随机变化有时可能会导致难以区分目标反射的信号和噪声尖峰。在图 10-13 中，接收机输出端的平均噪声功率电平用 $P_{no} = G_{rec}P_{ni}$ 表示，其中 G_{rec} 是接收机增益， P_{ni} 是接收机输入端子的噪声电平。功率水平 P_{r1} 和 P_{r2} 代表雷达观测到的两个目标的回波。由于噪声的随机性，需要设置一个阈值水平 P_{min} 来进行检测。对于图 10-13 所示的阈值级别 1，雷达将产生两个目标的存在，但它也会检测到误报。发生这种情况的几率称为误报概率。另一方面，如果将阈值级别提高到级别 2 以避免误报，雷达将不会检测到第一个目标的存在。雷达检测目标存在的能力由检测概率来表征。因此，相对于平均噪声水平的阈值信号水平的设置是基于权衡两种概率的折衷而进行的。

为了将噪声水平保持在最低水平，接收器的设计使其带宽 B 的宽度仅足以通过接收到的脉冲中包含的大部分能量。这种设计称为匹配滤波器，要求 B 等于脉冲长度 τ (的倒数，即 $B = 1/\tau$)。因此，对于匹配滤波器接收器，方程：(10.10) 变为

$$P_{ni} = KT_{sys}B = \frac{KT_{sys}}{\tau}. \quad (10.18)$$

雷达接收到的信号功率 P_r 通过雷达方程与发射功率电平 P_t 相关。我们将首先针对双基地雷达配置的一般情况推导雷达方程，其中发射机和接收机不一定位于同一位置，然后将结果专门针对发射机和接收机位于同一位置的单基地雷达情况。在图 10-14 中，目标距离发射机的距离为 R_t ，距离接收机的距离为 R_r 。照射目标的功率密度由下式给出

$$S_t = \frac{P_t}{4\pi R_t^2} G_t \quad (\text{W/m}^2), \quad (10.19)$$

其中 $(P_t/4\pi R_t^2)$ 表示各向同性辐射器辐射的功率密度， G_t 是各向同性辐射器的增益



Figure 10-13 The output of a radar receiver as a function of time.



Figure 10-14 Bistatic radar system viewing a target with radar cross section (RCS) σ_t .

transmitting antenna in the direction of the target. The target is characterized by a **radar cross section (RCS)** σ_t (m^2), defined so that the power intercepted and then reradiated by the target is

$$P_{\text{rer}} = S_t \sigma_t = \frac{P_t G_t \sigma_t}{4\pi R_t^2} \quad (\text{W}). \quad (10.20)$$

This reradiated power spreads out over a spherical surface, resulting in a power density S_r incident upon the receiving radar antenna. Hence,

$$S_r = \frac{P_{\text{rer}}}{4\pi R_r^2} = \frac{P_t G_t \sigma_t}{(4\pi R_t R_r)^2} \quad (\text{W}/\text{m}^2). \quad (10.21)$$

With an effective area A_r and radiation efficiency ξ_r , the receiving radar antenna intercepts and delivers (to the receiver)

power P_r given by

$$P_r = \xi_r A_r S_r = \frac{P_t G_t \xi_r A_r \sigma_t}{(4\pi R_t R_r)^2} = \frac{P_t G_t G_r \lambda^2 \sigma_t}{(4\pi)^3 R_t^2 R_r^2}, \quad (10.22)$$

where we have used Eqs. (9.29) and (9.64) to relate the effective area of the receiving antenna, A_r , to its gain G_r . For a monostatic antenna that uses the same antenna for the transmit and receive functions, $G_t = G_r = G$ and $R_t = R_r = R$. Hence,

$$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}. \quad \text{(radar equation)} \quad (10.23)$$

Unlike the one-way communication system for which the dependence on R is as $1/R^2$, the range dependence given by



图 10-13 雷达接收器的输出与时间的函数关系。



图10 -14 双基地雷达系统用雷达截面观察目标 (RCS) σ_t 。

发射天线指向目标方向。目标由雷达横截面 (RCS) σ_t (m^2) 来表征，定义为目标拦截然后再辐射的功率为

$$P_{\text{rer}} = S_t \sigma_t = \frac{P_t G_t \sigma_t}{4\pi R_t^2} \quad (\text{W}). \quad (10.20)$$

这种再辐射功率在球面上扩散，导致接收雷达天线上的功率密度 S_r 。因此，

$$S_r = \frac{P_{\text{rer}}}{4\pi R_r^2} = \frac{P_t G_t \sigma_t}{(4\pi R_t R_r)^2} \quad (\text{W}/\text{m}^2). \quad (10.21)$$

凭借有效面积 A_r 和辐射效率 ξ_r ，接收雷达天线拦截并传递 (至接收器)

功率 P_r 由下式给出

$$P_r = \xi_r A_r S_r = \frac{P_t G_t \xi_r A_r \sigma_t}{(4\pi R_t R_r)^2} = \frac{P_t G_t G_r \lambda^2 \sigma_t}{(4\pi)^3 R_t^2 R_r^2}, \quad (10.22)$$

我们使用等式的地方。 (9.29) 和 (9.64) 将接收天线的有效面积 A_r 与其增益 G_r 联系起来。对于使用同一天线实现发射和接收功能的单基地天线， $G_t = G_r = G$ 和 $R_t = R_r = R$ 。因此，

$$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}. \quad \text{(radar equation)} \quad (10.23)$$

与对 R 的依赖性为 $1/R^2$ 的单向通信系统不同，范围依赖性由下式给出：

the **radar equation** goes as $1/R^4$, which is the product of two one-way propagation processes.

The detection process may be based on the echo from a single pulse or on the addition (integration) of echoes from several pulses. We will consider only the single-pulse case here. A target is said to be detectable if its echo signal power P_r exceeds $P_{r_{\min}}$, which is the **threshold detection level** indicated in Fig. 10-13. The **maximum detectable range** $R_{\max}$ is the range beyond which the target cannot be detected, corresponding to the range at which $P_r = P_{r_{\min}}$ in Eq. (10.23). Thus,

$$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 P_{r_{\min}}} \right]^{1/4}. \quad (10.24)$$

The signal-to-noise ratio is equal to the ratio of the received signal power P_r to the mean input noise power P_{ni} given by Eq. (10.18):

$$S_n = \frac{P_r}{P_{ni}} = \frac{P_r \tau}{KT_{\text{sys}}}, \quad (10.25)$$

and the **minimum signal-to-noise ratio** $S_{\min}$ corresponds to when $P_r = P_{r_{\min}}$:

$$S_{\min} = \frac{P_{r_{\min}} \tau}{KT_{\text{sys}}}. \quad (10.26)$$

Use of Eq. (10.26) in Eq. (10.24) gives

$$R_{\max} = \left[\frac{P_t \tau G^2 \lambda^2 \sigma_t}{(4\pi)^3 KT_{\text{sys}} S_{\min}} \right]^{1/4}. \quad (10.27)$$

The product $P_t \tau$ is equal to the energy of the transmitted pulse. Hence, according to Eq. (10.27), *it is the energy of the transmitted pulse rather than the transmitter power level alone that determines the maximum detectable range*. A high-power narrow pulse and an equal-energy, low-power long pulse will yield the same radar performance as far as maximum detectable range is concerned. However, the range-resolution capability of the long pulse is much poorer than that of the short pulse [see Eq. (10.16)].

The maximum detectable range $R_{\max}$ also can be increased by improving the signal-to-noise ratio. This can be accomplished by integrating the echoes from multiple pulses in order to increase the total amount of energy received from the target. The number of pulses available for integration over a specified integration time is proportional to the PRF. Hence, from the standpoint of maximizing target detection, it is advantageous to use as high a PRF as allowed by other considerations.

10-7 Doppler Radar

The Doppler effect is a shift in the frequency of a wave caused by the motion of the transmitting source, the reflecting object,

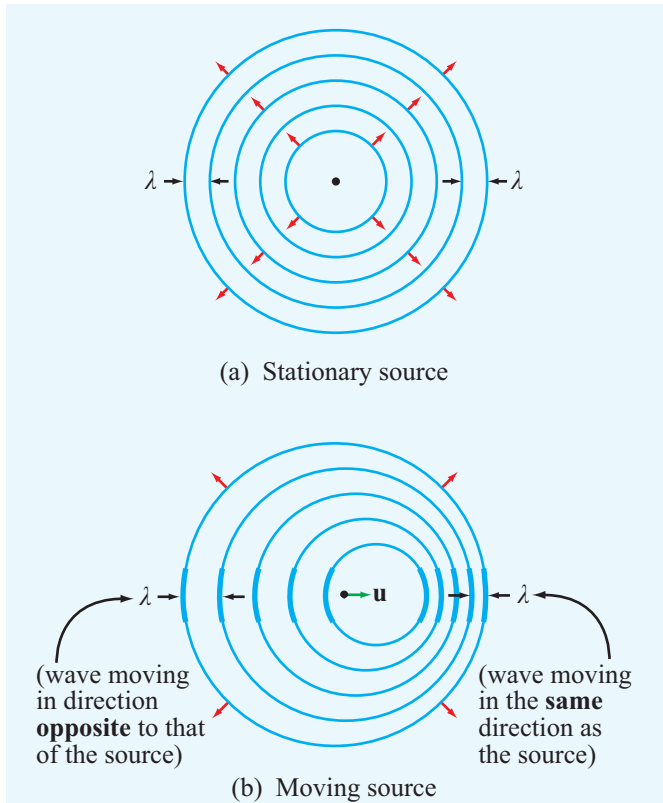


Figure 10-15 A wave radiated from a point source when (a) stationary and (b) moving. The wave is compressed in the direction of motion, spread out in the opposite direction, and unaffected in the direction normal to motion.

or the receiving system. As illustrated in Fig. 10-15, a wave radiated by a stationary isotropic point source forms equally spaced concentric circles as a function of time travel from the source. In contrast, a wave radiated by a moving source is compressed in the direction of motion and is spread out in the opposite direction. Compressing a wave shortens its wavelength, which is equivalent to increasing its frequency. Conversely, spreading it out decreases its frequency. The change in frequency is called the **Doppler frequency shift** f_d . That is, if f_t is the frequency of the wave radiated by the moving source, the frequency f_r of the wave that would be observed by a stationary receiver is

$$f_r = f_t + f_d. \quad (10.28)$$

The magnitude and sign of f_d depend on the direction of the velocity vector relative to the direction of the range vector connecting the source to the receiver.

Consider a source transmitting an electromagnetic wave

雷达方程为 $1/R^4$, 它是两个单向传播过程的乘积。

检测过程可以基于来自单个脉冲的回波或基于来自多个脉冲的回波的相加 (积分)。我们在这里仅考虑单脉冲情况。如果目标的回波信号功率 P_r 超过 $P_{r_{\min}}$ (图 10-13 所示的阈值检测水平), 则称目标是可检测的。最大可检测范围 $R_{\max}$ 是超过该范围就无法检测到目标的范围, 对应于式(1)中 $P_r = P_{r_{\min}}$ 的范围。 (10.23)。因此,

$$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 P_{r_{\min}}} \right]^{1/4}. \quad (10.24)$$

信噪比等于接收信号功率 P_r 与等式 1 给出的平均输入噪声功率 P_{ni} 之比。 (10.18) :

$$S_n = \frac{P_r}{P_{ni}} = \frac{P_r \tau}{KT_{\text{sys}}}, \quad (10.25)$$

而最小信噪比 $S_{\min}$ 对应于当 $P_r = P_{r_{\min}}$ 时:

$$S_{\min} = \frac{P_{r_{\min}} \tau}{KT_{\text{sys}}}. \quad (10.26)$$

使用等式。 (10.26) 式(10.24) 给出

$$R_{\max} = \left[\frac{P_t \tau G^2 \lambda^2 \sigma_t}{(4\pi)^3 KT_{\text{sys}} S_{\min}} \right]^{1/4}. \quad (10.27)$$

乘积 $P_t \tau$ 等于发射脉冲的能量。因此, 根据等式。根据方程 (10.27), 确定最大可检测范围的是发射脉冲的能量, 而不是发射器的功率电平。就最大可探测范围而言, 高功率窄脉冲和等能量、低功率长脉冲将产生相同的雷达性能。然而, 长脉冲的距离分辨率能力比短脉冲差得多[见方程1]。 (10.16)]。

最大可检测范围 $R_{\max}$ 也可以通过提高信噪比来增加。这可以通过对多个脉冲的回波进行积分来实现, 以增加从目标接收到的能量总量。在指定的积分时间内可用于积分的脉冲数量与 PRF 成正比。因此, 从最大化目标检测的角度来看, 使用其他考虑因素允许的尽可能高的 PRF 是有利的。

10-7 多普勒雷达

多普勒效应是由发射源、反射物体的运动引起的波频率变化,

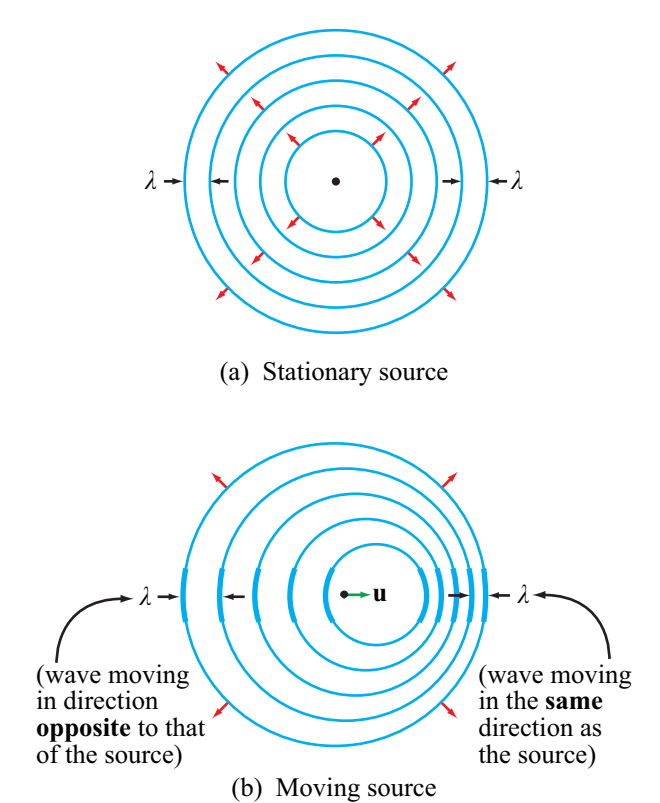


图 10-15 当 (a) 静止和 (b) 移动时从点源辐射的波。波在运动方向上被压缩, 在相反方向上展开, 并且在垂直于运动的方向上不受影响。

或接收系统。如图 10-15 所示, 由静止各向同性点源辐射的波形成等距同心圆, 作为从源传播的时间的函数。相反, 移动源辐射的波在运动方向上被压缩, 并在相反方向上扩散。压缩波会缩短其波长, 相当于增加其频率。相反, 将其展开会降低其频率。频率的变化称为多普勒频移 f_d 。也就是说, 如果 f_t 是移动源辐射的波的频率, 则静止接收器观测到的波的频率 f_r 为

$$f_r = f_t + f_d. \quad (10.28)$$

f_d 的大小和符号取决于速度矢量相对于连接源和接收器的距离矢量方向的方向。

考虑发射电磁波的源